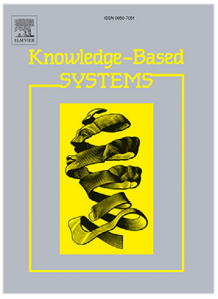
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Madhavi, Thipendra P. Singh, Shailendra Pratap Singh

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ST-APCAR: A Novel Approach to Enhance Swin Transformer for Leaf Disease Classification on Heterogeneous Datasets

Madhavi*[a], Thipendra P Singh[b], Shailendra Pratap Singh[c]

[a]*School of Computer Science Engineering & Technology, Bennett University, TechZone 2, Greater Noida, 201310, U.P., India*
[b]*Chandigarh University Uttar Pradesh, Lucknow-Kanpur Highway-27, Unnao, 209859, U.P., India*
[c]*Madan Mohan Malaviya University of Technology, Gorakhpur, 273010, U.P., India*

Abstract

The agriculture domain contains various subcategories, and leaf diseases are among the most dominant and impactful challenges. The identification of leaf diseases faces challenges due to its complex disease characteristics, similar appearances among multiple classes, and the sensitivity of models relying on attention mechanisms. In this work, we introduce a Swin Transformer-based learning framework (ST-APCAR) that systematically enhances classification performance through four complementary mechanisms: *Adaptive Learning Rate Scheduling*, *Pathway Regularization*, *Curriculum Learning*, and *Attention Norm Regularization*. The proposed framework was developed as a coordinated training strategy in which each component addresses a specific weakness of transformer learning, including optimization instability, difficulty in progressive representation learning, unbalanced attention responses, and inefficient feature propagation for leaf disease classification. It investigates the effect of each proposed strategy independently to reveal its contribution to performance improvement and also validates the full ST-APCAR framework on two datasets using a common experimental setting. In the Agriculture Crops dataset, the proposed methods achieve 99.88% accuracy for Adaptive Learning Scheduling, 92.76% for Curriculum Learning, 91.92% for Pathway Regularization, and 92.76% for Attention Norm Regularization. The ST-APCAR architecture performed best with an accuracy of 99.92%. On the PlantDisease dataset, the models based on Adaptive Learning Scheduling, Curriculum Learning, Pathway Regularization, and Attention Norm Regularization achieve accuracy of 99.89%, 94.93%, 93.52%, and 96.92%, respectively. The ST-APCAR framework achieved 99.93% accuracy. Further, the superpixel-based score-drop analysis and Local Interpretable Model-agnostic Explanations (LIME) were used to analyze model predictions. These techniques highlight supporting and counter-evidence regions that drive classification decisions, and also provide both qualitative visualization and faithfulness assessment. The findings show that each proposed component improves the Swin Transformer, while their unified integration provides the most effective and consistent performance for leaf disease classification.

Keywords: Leaf disease classification, Swin Transformer, Attention norm regularization, Curriculum learning, Adaptive learning, Pathway regularization,

1. Introduction

Agriculture is a vast domain encompassing various categories of plants and leaves. It is the foundation for food supply and economic growth, which in turn supports millions in their livelihoods. The greatest and most prevalent challenge in agriculture is the identification of leaf disease, as it significantly affects productivity. Recent technological advancements in machine learning and deep learning models provide an effective solution for disease identification. The Vision Transformers are among the latest advances in deep learning that have demonstrated their potential for complex image-related tasks. The Swin Transformer is a variant of the standard Vision Transformer that provides hierarchical processing of an image and gives more attention to localized features. The transformer models can define relationships in data without being constrained by sequential processing, which makes them particularly suitable for image-based tasks.

The transformer focuses on the diseased areas of the leaf and captures small features that help classify and detect leaf disease at different stages. Training such models is costly due to the size of the dataset, its complexities, model size, and time required for training. Leaf disease identification has advanced from conventional CNN-based models to transformer-based and hybrid frameworks because the feature extraction and classification depend on efficient training. The authors [1] proposed a hierarchical framework that combines Residual Vision Transformer with ResNet9 to capture discriminative disease features, also reducing trainable parameters and computational cost. The study, with 97.5% classification performance, shows that hierarchical transformer-based learning is effective for plant pathology. The work mainly focuses on architectural improvement but fails in model generalization under diverse conditions. Similarly, the authors [2] propose adapters, overlapping patch embedding, and Softmax loss to show a parameter-efficient transformer model. The model with the best accuracy, 98.67%, also showed strong performance in few-shot settings with fewer trainable parameters. Even if the model shows improved parameter efficiency, it still highly depends on dataset

quality and tuning. To improve the previous research, the authors [3] present a hybrid CNN–ViT framework, which combines local feature extraction of CNNs with the global representation ability of Vision Transformers. The approach finally showed an accuracy of 99.24% with good multi-class disease detection performance, but the overall complexity and training cost increased. In a related direction, the authors [4] presented a multi-scale feature-fusion adaptive deep network MFF-ADNet for identifying different plant and leaf diseases.

The analysis showed an improved median value that contains features from multiple scales and improves disease-type recognition. However, the method mainly emphasizes classification accuracy, with less attention to computational efficiency. In addition to architectural design, recent studies suggest that training strategy is also crucial for improving classification performance. The authors [5] showed the improved training stability and efficiency achieved in physics-informed neural networks (PINNs) and image-classification tasks using a dynamic learning-rate schedule. However, the work does not specifically validate the approach on plant disease datasets. Similarly, the authors [6] proposed an AdaGL adaptive optimizer that integrates long-term and short-term gradient information for real-time update control. The authors reported faster convergence, stable training, and good generalization across deep-learning tasks with 88.79 % on benzodiazepine receptor(BZR). However, its direct suitability for plant and leaf disease classification is unexplored.

In the same optimization context, the authors [7] presented a layer-wise adaptive learning-rate method, AdaLip, for stochastic optimization, which improves convergence speed and robustness, and also reduces sensitivity to the initial global learning rate. Curriculum-based optimization shows another useful perspective for model training, where the authors [8] presented Learning Rate Curriculum (LeRaC) in which the different layers were assigned different learning rates during early training to improve convergence and generalization. The curriculum-based learning-rate control, with the mean average precision (mAP) of 0.846, can benefit different architectures. In a related idea, the authors [9] applied a two-step curriculum learning strategy for chest X-ray lesion classification, where training progresses from lesion patches to full images. Although this work is from medical imaging, it suggests that progressive learning strategies may also help plant leaf disease classification. The approach is not validated in a different application domain, so direct transfer to agricultural datasets may not always be straightforward.

For the proposed research, the Swin Transformer was chosen as the architecture because of its hierarchical feature representation, efficient computation, and capability to learn local and cross-window context. These characteristics make Swin Transformer appropriate for leaf disease images that have different symptom scales and distributions. The research proposes a new approach of four techniques: Attention Norm Regularization, Pathway Regularization, Curriculum learning, and Adaptive learning rate scheduling, where each technique plays a significant role in leaf disease identification and classification. In this research, the Attention Normalization technique is used to normalize attention scores in transformer blocks, thus, the model does not focus too much on the same feature or pattern during training. To support this, the Pathway Regularization encourages the model to explore different computational pathways through its layers. It doesn't focus more on an extremely narrow collection of features, but generalizes better across different types of leaf conditions and with various variances. To further support this learning process, the curriculum learning procedure and the training data are slowly complicated to mimic human cognitive development. The technique keeps improving the learning by prioritizing easier instances early in the training phase, which results in a smoother learning gradient. In addition, adaptive learning rate scheduling adjusts the learning rate during training based on the model's progress. It avoids local minima and adjusts the training process more precisely by preventing them. Altogether, these techniques are used to make the Swin Transformer model more stable, robust, and effective for leaf disease classification. In this research, the performance of the proposed work is tested on a comprehensive leaf image dataset and compared with existing methods. The objective is to identify whether the proposed approach can improve classification accuracy under different disease conditions.

1.1. Research Objective

The *objective* of the research are as follows:

- Attention Norm Regularization manages the direct output from the attention mechanism, which affects the way the model assigns its focus across input features. Attention Norm Regularization was developed within the Swin Transformer architecture, where the regularization technique focuses on normalizing attention mechanisms across different layers.
- Pathway Regularization analyses the multiple computation paths in the transformer. The main task is to balance the dependency of the model across different paths. This makes it resilient and generalized to different representations and orientations of leaf diseases. It also helps to prevent the model from becoming dependent on specific paths within the network.
- Curriculum Learning for Attention continuously improves the model's exposure to complexity. This can be specifically useful in preventing early overfitting and easing the learning of complex patterns.
- Adaptive Learning Rate Scheduling based on the observed behavior of the attention mechanism dynamically adjusts the learning rate, which helps to stabilize training and improve model convergence.

1.2. Motivation for Proposed Work

The Swin Transformer is recognized to be highly efficient at solving problems related to vision using hierarchical representation learning and shifted window-based attention mechanisms. However, there are several drawbacks in the use of this model that hinder the efficiency with which it solves visual problems, such as identifying leaf diseases.

- The attention mechanism used by the standard approach tends to give highly localized and concentrated attention distributions, focusing on a few disease-dominated areas only, neglecting other useful spatial information.
- The Swin Transformer model does not explicitly seek diversity within the computational pathways, making the model overly dependent on a small proportion of feature representations.
- There is no learning progression structure in the design of the model, making it difficult for the model to learn features progressively, hence leading to instabilities.
- There is no coordination between attention evolution and the optimization processes of the model, making learning and convergence less efficient and slower.

These limitations motivate the need for a structured enhancement framework that can regulate attention distribution, encourage pathway diversity, show progressive learning, and adapt the optimization process dynamically.

1.3. Author Contribution

To address the *limitations*, we introduce four complementary strategies.

- Attention Norm Regularization: It regulates the attention distribution and controls excessive concentration on a small number of tokens.
- Pathway Regularization: It encourages the use of multiple computational pathways and improves representative diversity and robustness.
- Adaptive Learning Rate Scheduling: Dynamically adjusts the learning rate according to attention statistics, which allows the optimization process to respond to the model's internal learning state.
- Curriculum Learning: Stabilizes and enables smoother optimization of the training process by slowly increasing the learning complexity over epochs.

The combined effect on attention-driven transformer architectures remains underexplored in the prior studies. There is a lack of a structured framework that jointly addresses attention stability, feature diversity, progressive learning, and adaptive optimization. This motivates the proposed ST-APCAR framework. The proposed approaches do not mimic the Swin Transformer model; rather, they aim to improve the baseline model's performance in training, robustness, and generalization for classifying leaf diseases. The four proposed enhancements have been considered. Still, they have not been chosen randomly, since they provide complementary methods to solve attention distribution, representation diversity, optimization stability, and convergence adaptability issues in Swin Transformer. Motivation of the problem-driven enhancements over the baseline Swin Transformer for leaf disease classification is provided in Table 1.

The proposed ST-APCAR framework does not aim to present a new transformer backbone. Rather, the framework is built upon the Swin Transformer and presents an attention-guided training and regularization framework for leaf disease classification. The curriculum learning, adaptive learning-rate scheduling, and regularization are from prior learning paradigms. ST-APCAR uses them differently due to its internal attention behavior. The framework comprises four complementary attention-guided components: deviation-based Attention Norm Regularization, pathway-level consistency regularization, curriculum-scaled attention using an epoch-dependent $\beta(t)$, and attention-dispersion-driven adaptive learning-rate scheduling. Hence, the contribution of ST-APCAR lies in the mathematical adaptation and coordinated integration of these strategies in Swin Transformer training. The framework is not replacing or redesigning the original transformer mechanism.

This research paper focuses mainly on SDG 2 (Zero Hunger) and SDG 12 (Responsible Consumption and Production), because the correct detection and classification of diseases help prevent crop loss and improve food security through efficient farming. The sections of this research paper are structured as follows: Section 2 provides an overview of the current approaches. Section 5 presents the proposed framework for disease identification and classification. The results of the techniques are presented in Sections 6 & 7. Section 9 concludes the paper with future work.

2. Related Work

Various researchers have proposed techniques for leaf disease identification and classification using various Machine learning and Deep learning methods. In this direction, the authors [10, 11] in the survey show the strengths of transformers in capturing long-range dependencies and learning rich feature representations. Focusing on the role of transformers, the authors also recognized hierarchical vision transformers with Swin Transformer. These are efficient alternatives to vision transformers, but the survey does not focus on disease classification, dataset imbalance, and subtle symptom differences. The authors [12] in their research presented a flexible vision transformer. Here, the Swin Transformer uses a hierarchical structure with shifted window-based self-attention. The study reduces computational cost and improves scalability for high-resolution images, and also shows a margin of +2.7 box AP compared to traditional models. This work is strong in image classification and dense prediction, but lacks fine-grained plant disease variation.

Similar to previous research, the authors [13] also present a hierarchical vision transformer model for classifying leaf diseases with transfer learning. The research uses 22 different types of leaf images with a multi-head attention module and a multi-layer perceptron to improve contextual understanding of the images. To compress training, the author uses ImageNet-pretrained weights and transfer learning and shows 86.43% classification accuracy. The model's lack of generalizability and robustness under broader real-field conditions. To improve the research and to develop a dual-track feature-fusion architecture which can classify grape leaf diseases, the au-

Table 1: Problem-driven motivation of the proposed enhancements over the baseline Swin Transformer for leaf disease classification.

Issue in baseline Swin Transformer	Importannt for leaf disease classification	Scientific motivation	Proposed enhancement
Self-attention focuses on certain visual regions due to their prominent characteristics, especially when there are lesions with high contrast that receive more attention.	Leaf diseases have features that are difficult to detect, such as spots, color change on the borders, deformations in the veins, and unusual textures. When diagnostic patterns are overly focused on a few regions, they may be missed in other parts of the leaves.	Fine-grained image analysis requires balanced attention since disease areas in the images are usually scarce, localized, and not uniformly spread throughout the image. There needs to be a way of preventing attention failure and focusing on the relevant regions in the leaves.	**Attention Norm Regularization** is proposed to avoid extreme concentration of attention and allow for more balanced attention.
The learned feature representations can be biased towards a subset of internal pathways, which leads to poor diversity in discriminative information learned.	Disease classes usually have strong similarities within the class, but they lack separability between them. Multiple features like color, texture, lesion boundary, and shape play an important role in distinguishing disease classes.	Diversity is very important when differentiating visually similar disease classes. Pathways are stimulated to take advantage of different disease characteristics, rather than focusing on one apparent path.	**Pathway regularization** minimizes the dependency on certain pathways for feature representations and learns a richer and diverse set of representations.
Learning can be unstable in early phases of training due to the large variability of the appearance of diseases, the background clutter, and the difficulty variations among the classes.	Leaf images include varying levels of symptom severity, changes in illumination conditions, partially hidden samples, and complex backgrounds.	Learning all images with varying levels of difficulty in one phase would not result in a faster convergence, leading to an optimization focused on simpler visual cues.	**Curriculum Learning** is used to increase the learning difficulty and make learning more stable, and prevent overfitting to simpler patterns.
Fixed learning rates cannot always shift attention and feature-stabilization behavior during training.	During leaf diseases classification, the training process may experience some variation at different stages, since in the initial stages, the goal is the recognition of basic patterns, while in the latter stages, the objective is the identification of more subtle patterns.	The adaptation approach of optimization should be considered in the case where the training procedure is heterogeneous and where representation changes are taking place.	**Adaptive Learning Rate Schedule** is chosen for controlling parameter updates according to the training progress and learning dynamics.
While the design of Swin Transformer offers hierarchical representations and a window-based self-attention mechanism, it fails to provide any explicit consideration toward interpretability, attention diversity, or adaptation during training in order to learn from disease-specific visual variations.	Identification of plant diseases is a highly fine-grained task requiring the model to capture both minor features of symptoms and lesions as well as the ability to adapt to large visual differences.	As such, a simple Swin Transformer becomes an ideal backbone network, but further optimization is required to enhance stability and generalizability of the model in the context of disease identification within agriculture.	In this work, we develop an augmented version of the **Swin Transformer** utilizing four auxiliary techniques: **Attention Norm Regularization**, **Pathway Regularization**, **Curriculum Learning**, and **Adaptive Learning Rate Scheduling**.

thors [14] combine a Group Shuffle Residual DeformNet and a Swin transformer. With an outstanding accuracy 98.6% on the PlantVillage dataset. Further to optimize feature interaction and sampling, the Triplet Attention and deformable convolution are utilized. This method offers the automation process for classifying grape leaf diseases. Similar to the research, the authors [15] proposed an ensemble of Swin transformers and residual convolutional neural networks to provide early and precise diagnosis of plant diseases. This approach enhances the performance of plant disease classification due to the efficiency of local and global feature extraction. However, the model was mainly validated on the controlled PlantVillage dataset, so its robustness and generalizability to real-field plant disease images remain uncertain. The authors [16] use a hybrid feature fusion model based on MobileNetV3, Swin Transformer, and a variational autoencoder for coffee leaf diseases classification on the RoCoLe dataset. Based on early and late fusion techniques, this approach scores 84.29% accuracy. The effective integration of features extracted from various frameworks, the approach has performed more efficiently compared to the traditional methods and has the potential of achieving better effectiveness when applied to the real-world setting. Compared to the previous research, the authors [17] discuss a novel segmentation model based on Transformer architecture called AISOA-SSformer. In this model, the integration of salient feature attention mechanism, annealing integrated sparrow optimization approach, and sparse global-update perceptron helped achieve an MIoU and Dice coefficient of 83.1% and 80.3%, respectively. Whereas the Squeeze-and-Excitation Vision Transformer (SE-ViT), integrates ViT and channel attention with ResNet for plant disease classification, proposed by authors [18]. Where SEViT achieved 88.34% test accuracy for large-scale, fine-grained plant disease classification, improving on the baseline by 5.15%. The result shows that fine-grained discrimination remains challenging, especially when many disease classes have highly similar visual symptoms.

In identifying plant diseases under difficult circumstances, the authors [30] examine attention mechanisms and highlight how classification models are more reliable and accurate. Attention-based mechanisms, including Convolutional Block Attention Module, Squeeze, and Excitation, are used to examine performance improvement through models such as MobileNetV2, EfficientNetV2, and ShuffleNetV2. As a consequence, the highest accuracy score is 99.96%, thereby validating the hypothesis that incorporating attention mechanisms enhances existing deep learning networks. For detecting plant diseases on sugarcane leaves, the authors [31] propose SLViT, a novel combination of a vision transformer model. On the PlantVillage dataset, SLViT yields an accuracy of 98.84% with superior computation speed and model size when compared to other previous models. On the novel SLD10k dataset, SLViT exceeds the performance of MobileNetV3 Small with an additional 1.87% accuracy and 66.3% model size. The model has practical applications in diagnosing agricultural diseases due to its efficiency in the field and its patience with background noise. In contrast to the research, the authors [32] propose an effective Curriculum learning approach, which involves learning simpler patterns first and then moving onto more complex ones to avoid the computational complexity associated with the conventional approaches. This can be achieved by beginning

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Table 2: Critical synthesis and quantitative summary of recent studies in plant disease classification.

✓: explicitly addressed; ×: not addressed; △: partially / indirectly addressed.

Reference	Disease	Dataset	Model	Metric (%)	Robus Generaliz	Explain-ability	Realistic Field Data	Observed Gap
[19]	Potato leaf disease	Potato leaf image dataset	Swin Transformer	99.70	△	×	△	Strong classification accuracy, but emphasis is mainly on performance; limited model interpretability and no explicit uncertainty or cross-domain robustness analysis.
[20]	Potato disease	PlantVillage	MobOca_Net (based on MobileNetV2)	97.33	×	×	×	Lightweight and accurate, but evaluated on a controlled benchmark dataset; limited evidence of explainability and real-field generalization.
[21]	Abnormal potato plant detection	Field video images of potato plants	Fast R-CNN, YOLOv3	96.70	✓	×	△	Detection-oriented framework improves practical relevance, yet explainability and disease-specific decision transparency remain insufficiently addressed.
[22]	Cassava leaf diseases (CBB, CBSD, CGM, CMD, healthy)	Kaggle Competition 2020	EfficientNetB3	83.30	△	×	△	Addresses an important real crop problem, but comparatively lower performance and lack of interpretability reduce reliability for practical deployment.
[23]	Cassava leaf diseases	NaCRRI	CNN	85.38	×	×	△	Conventional CNN baseline provides moderate accuracy only; it lacks explainability, robustness validation, and broader disease coverage.
[24]	Olive leaf spot diseases	Collected dataset	Texture analysis + SVM	97.12	△	×	△	Good performance on collected data, but generalization to larger and more diverse field environments is not clearly established.
[25]	Potato plant leaf disease	PlantVillage	EfficientPNet	98.12	△	✓	△	Incorporates visual explanation, but robustness across domains and deployment on highly variable field data remains limited.
[26]	Cassava leaf disease	Cassava Leaf Disease Dataset	FormerLeaf	Acc. improvement of about 3%; speed improvement of 15%	△	×	×	Efficiency gains are reported, but dataset transparency, explainability, and practical field validation are insufficiently discussed.
[27]	Six species of flying insects	DIRT olive fruit fly dataset	Modified YOLO	96.68	△	×	✓	Useful for agricultural pest monitoring, but focuses on insect detection rather than plant disease diagnosis and does not address interpretability.
[28]	Maize leaf diseases	18,888 healthy and diseased leaf images	VGG16, ResNet50, InceptionV3, Xception	99.0	△	×	△	Very high accuracy is reported, yet the study remains primarily performance-centric without explicit explainability or robustness analysis.
[29]	Downy mildew and powdery mildew	33 grape leaf images	K-means clustering & neural network	100	×	×	×	Early classical approach with extremely limited sample size; results are unlikely to reflect real-world variability or scalable deployment.

with learning simpler patterns, like low-frequency image patterns. Learning drastically shortens training time and costs without compromising accuracy. This approach works well in situations involving self-supervised learning as well. The research reduced wall-time training cost by more than 1.5× on ImageNet-1K/22K without sacrificing accuracy. However, the method is only validated mainly on large-scale benchmark datasets. Similar to the research, the authors [33] represent an extreme curriculum learning approach to decrease the training times of computer vision models without compromising accuracy. It uses a soft-selection function that starts with basic patterns and slowly uncovers models to more complex ones inside each training sample. This method shows significant conservation in training times and is computationally efficient for a wide range of models. Whereas the authors [34] propose a unique Graph Transformer-based semi-supervised node classification algorithm. It uses a Local Graph Transformer as the teacher and a Global Graph Transformer as the student. The concept uses an attention distillation method, which is inspired by the curriculum. To enhance distillation, this approach adds an out-of-class entropy and curriculum mechanism, showing Node classification performance of 84.70, resulting in better performance on a variety of graph datasets. The authors [35] propose a GREnet model for 2-D medical image segmentation, which combines curriculum learning in a novel way by handling the segmentation job in a temporal sequence that goes from basic to more complicated circumstances, by concentrating on more manageable portions of the data. The growing complexity in segmentation tasks by the curriculum learning framework provides more accurate and effective learning. The authors [36] for medical image segmentation provide a unique domain adaptation approach that integrates curriculum learning with Fourier Transform amplitude modifications. It is slowly adding multiple features to the images from the source domain, raising the complexity of training samples.

The study also merges the chained augmentation approaches, which enhanced resilience and adaptation, boosting performance in several datasets, and shows promising results in domain-adaptive classification tasks. The authors [37] present the MSST-Net model as an improved multi-scale adaptive network for remote sensing image segmentation, which is based on Swin Transformers. It integrates different scale features for precise segmentation and uses self-attention processes for global feature recognition. The authors [38] offer instance-

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Figure 1: Data Augmentation for the AG dataset and PD dataset

level document segmentation called SwinDocSegmenter based on the Transformer. Focusing on low-level details and high-level features, it uses a unified encoder-decoder architecture with a Swin Transformer backbone to improve segmentation. A hybrid bipartite matching approach combines domain adaptation with contrastive denoising to improve the handling of underrepresented classes. Experiments conducted on various datasets show that SwinDocSegmenter performs better than current techniques, showing strong generalization abilities and efficiency in complex layout segmentation. The authors [39] present a transformer-based, self-adaptive model for reliable and accurate skin lesion segmentation. It uses a position-aware segmentation model to properly address skin lesions' spatial distribution and characteristics, increasing accuracy and decreasing mis-segmentations. This model shows 97.9%, 94.3%, and 95.7% accuracy on several datasets by utilizing a transformer architecture to facilitate multi-scale, positional feature extraction and optimization. The summary of recent work is mentioned in Table 2.

3. Dataset Acqusition

Agriculture Crops Dataset (AG dataset) chosen from Kaggle [40] consists of leaf images of various plant diseases like almond, banana, cardamom, Cherry, chili, clove, coconut, Coffee-plant, cotton, Cucumber, Fox_nut(Makhana), gram, jowar, jute, Lemon, maize, mustard-oil, Olive-tree, papaya, Pearl_millet(bajra), pineapple, rice, soybean, sugarcane, sunflower, tea, Tobacco-plant, tomato, vigna-radiati(Mung), and wheat. We have focused on all the classes for Training and testing purposes.

The performance testing is done on diseased region masses, structures, sizes, orientation, color fluctuations, clutter, blur, and intensity variations. For data augmentation, we used techniques like RGB shift, HSV adjustment, Channel shift, CLAHE, Random contrast, Random Gamma, Random brightness, blur, Median Blur, To Gray, and JPEG compression. As shown in Fig. 1. Plant Disease Dataset (PD Dataset)is taken

Table 3: Terminology & Notation used in Techniques

Symbol	Description
A	Attention weights
L1 or L2	Norm regularization
λ	Regularization coefficient.
p	Order of the norm (1 or 2).
L_reg	Regularization term.
δ	Tuning parameter.
L_{main}	Standard loss
$\beta(t)$	Scheduling parameter
k	Small constant
lr_0	Learning rate.
η	Sensitivity parameter.
std(·)	Standard deviation

from Kaggle [41], which consisted of various plant categories and diseases such as: Tomato Spider mites Two spotted spider mite, Tomato Septoria leaf spot, Tomato Leaf Mold, Tomato Late blight, Tomato healthy, Tomato Early blight, Tomato Bacterial spot, Tomato Tomato Yellow Leaf Curl Virus, Tomato Tomato mosaic virus, Tomato Target Spot. We have focused on all the classes for Training and testing purposes. The testing is done on diseased region masses, sizes, orientations, structures, color fluctuations, blur, clutter, and intensity variations. We perform data augmentation using techniques like RGB shift, Channel shift, CLAHE, Random contrast, Random Gamma, To Gray, Random brightness, HSV adjustment, blur, Median Blur, and JPEG compression, as shown in Fig. 1.

Class Distribution and Imbalance Handling

To assess the class imbalance, the class distributions of the datasets were analyzed. There were some minor variations in class frequencies across different disease categories, but no severe imbalance was observed. During dataset splitting, the Stratified sampling was used to ensure each subset maintains the original class distribution. To analyse the performance, F1-score and MCC metrics were used as these are less sensitive to class imbalance.

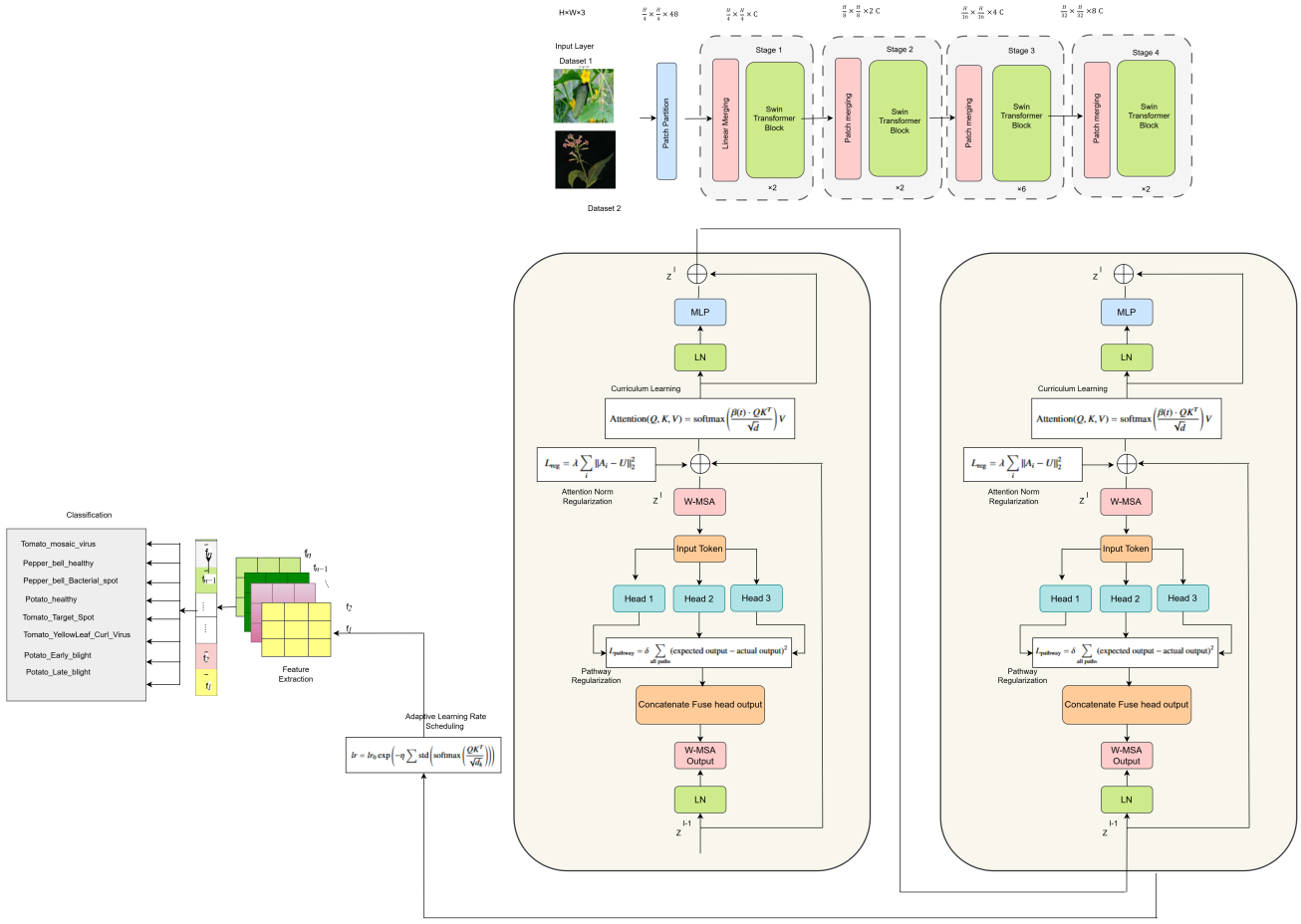


Figure 2: Swin Transformer architecture with Adaptive Learning Rate Scheduling, Pathway Regularization, Curriculum Learning, and Attention Norm Regularization

4. Swin Transformer

The Swin Transformer is a new hierarchical architecture known for its effectiveness and efficiency in a variety of visual tasks. It allows scaling across various image sizes and resolutions as shown in Fig. 2. The Swin Transformer was selected as the backbone architecture due to its hierarchical feature representation, computational efficiency, and its ability to capture both local and cross-window contextual information. These features are highly suitable for leaf disease images containing multi-scale and spatially distributed symptoms. Unlike the existing CNN architectures that rely mostly on the local receptive field approach, the Swin Transformer is unique as it allows modeling of more complicated contextual relationships without compromising its scalability properties. This makes the architecture applicable to leaf disease classification, where disease symptoms may be present at various spatial scales and in diverse visual environments.

The following are the necessary actions in the Swin Transformer model:

- **Patch Partitioning:** The image has patches each measuring 4 by 4 pixels. Each patch is referred to as a "token," and the number of tokens is based on the size of the patch and the image.
- **Embedding Tokens:** Every patch is flattened and then linearly transformed into an embedding space. A set of patch embeddings that resemble NLP word embeddings is generated during the process.
- **Window-based Self-attention:** The transformers calculate self-attention globally, whereas the Swin Transformer limits self-attention to small windows (7×7 patches).
- **Shifting Window Scheme:** Swin Transformer overcomes limitations by using a shifting window method for cross-window communication. In this technique, windows in alternate levels are shifted by a small number of pixels (half the window size) to expose later layers to a larger range of patches.
- **Hierarchical Feature Map:** The resolution of the feature maps gradually drops as the embedding dimensions grow

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in this pyramid-like architecture. This hierarchical structure enables efficient processing and multi-scale representation.

- **Feed-forward Layers:** Each transformer block has a feed-forward network that is applied differently to each token, in keeping with standard transformers.

- **Downsampling:** There is a patch merging layer used between different resolution levels. It doubles the feature dimensions and reduces the number of tokens by concatenating the features of neighboring patches (2x2) to form a single patch.

5. Proposed Work

In this section, we will discuss all four techniques, such as Attention Norm Regularization, Pathway Regularization, Curriculum Learning, and Adaptive Learning Rate Scheduling.

5.1. Attention Norm Regularization

Attention Norm Regularization helps manage the attention mechanism's direct outputs. It also ensures the model does not ignore or overemphasize certain features disproportionately. This led to more balanced learning and better generalization across different leaf diseases.

Base Self-Attention Formula: The standard self-attention mechanism in transformers can be expressed mathematically as follows:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{1}$$

Here, the Q, K, and V represent the query, key, and value matrices, respectively, and these were derived from the input data. The d_k by normalizing the dot products, controlling them from growing too large, and representing the dimensionality of the keys and queries. The Softmax function is applied in each row of the resulting matrix to turn logits into probabilities that sum to 1. The Attention Norm Regularization focuses on controlling the distribution of attention weights. This helps the model avoid overly focusing on a limited subset of features. To directly impose a penalty on the L1 or L2 norm of the softmax outputs, the formulation regularizes the deviation of the attention distribution from a reference distribution. Hence, it provides a meaningful optimization signal and encourages balanced attention allocation.

Softmax Application:

- It calculates the raw attention scores:

$$S = QK^T$$

- The scores normalize by $\sqrt{d_k}$:

$$S' = \frac{S}{\sqrt{d_k}}$$

Algorithm 1 Swin Transformer with Attention Norm Regularization

Require: Training dataset $D = \{(x_1, y_1), \ldots, (x_n, y_n)\}$, number of classes *num_classes*, model dimensions *dim*, number of attention heads *num_heads*, dropout rate *dropout*, learning rate *lr*, number of epochs E, batch size B, regularization weight λ
Ensure: Trained Swin Transformer model f with parameters θ

function AttentionNormRegularization(attention_weights, λ)
  $A \leftarrow$ softmax(*attention_weights*)
  $N \leftarrow$ number of tokens in each attention row
  $U \leftarrow [\frac{1}{N}, \frac{1}{N}, \ldots, \frac{1}{N}]$
  $L_{\text{reg}} \leftarrow \lambda \sum_i \|A_i - U\|_2^2$
  return L_{reg}
end function
function SwinTransformer(num_classes, dim, num_heads, dropout)
  patch_embedding $\leftarrow$ Initialize patch embedding layer
  transformer_blocks $\leftarrow$ Stack of Swin Transformer layers
  classifier $\leftarrow$ Linear(dim, num_classes)
  return Compound model of patch_embedding, transformer_blocks, and classifier
end function
model $\leftarrow$ SwinTransformer(num_classes, dim, num_heads, dropout)
model.to(device)
optimizer $\leftarrow$ Adam(model.parameters(), lr = *lr*)
criterion $\leftarrow$ CrossEntropyLoss()
for epoch = 1 to E **do**
  for each batch (x, y) in D **do**
    logits, attention_weights $\leftarrow$ model(x)
    $L_{\text{cls}} \leftarrow$ criterion(logits, y)
    $L_{\text{reg}} \leftarrow$ AttentionNormRegularization(*attention_weights*, λ)
    $L_{\text{total}} \leftarrow L_{\text{cls}} + L_{\text{reg}}$
    Backpropagate L_{total} and update model parameters
  end for
end for

Note: The attention weights are obtained from the curriculum-adjusted attention mechanism.

- The softmax is applied to these scores to get the attention weights:

$$A = \text{softmax}(S')$$

here each element of A now represents a probability.

By penalizing the attention weights norm, the model avoids overdependence on specific input. Hence, improve the generalization and reduce the chances of overfitting dominant features in the training data of the model. The λ shows the trade-off between the data and follows the regularization principle. Further, it is tuned based on validation performance. The regularization behavior is based on the choice of L1 and L2 norms. Here, the L1 norm advances sparsity in the attention weights and the L2 norm enables distributed weights.

Limitation of Direct Norm Regularization: A naive method of regularization involves applying L1 or L2 norms directly on the attention weights:

$$L_{\text{reg}} = \lambda \|A\|_p \tag{2}$$

Here, the $A = \text{softmax}(\cdot)$ and each row of A sums to one due to the normalization condition set by the softmax function. Therefore, the L1 norm does not provide any significant optimization signal and becomes constant, whereas the L2 norm stays bounded within a narrow range. This finally limits the effectiveness of such regularization.

Proposed Regularization Formulation:

To address this issue, we refine the regularization strategy. Here, we penalize the deviation of attention distributions from a reference distribution instead of their absolute magnitude.

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Let A_i denote the attention distribution corresponding to the i-th query, and let U represent a uniform distribution:

$$U = \left[\frac{1}{N}, \frac{1}{N}, \ldots, \frac{1}{N}\right] \tag{3}$$

Here, the number of tokens in the attention window is N and the Attention Norm Regularization is defined as:

$$L_{\text{reg}} = \lambda \sum_i \|A_i - U\|_2^2 \tag{4}$$

This technique encourages the attention mechanism to keep a balanced distribution across multiple tokens. It prevents excessive attention on specific regions and provides flexibility for task-specific feature learning. Unlike the direct norm regularization during training, this process provides a significant gradient signal that shapes the attention distribution. The model controls the variation from uniformity and improves robustness to noise, occlusion, and illumination changes by avoiding over-dependence on dominant features. The hyperparameter controls the stability of regularization λ, which balances task-specific learning and distributional regularization. Based on the validation performance, the optimal value of λ is selected.

5.2. *Pathway Regularization*

The objective of the Pathway Regularization technique is to diversify the dependence on various pathways within transformers. It controls the model from relying on any specific pathway and enhances its robustness and generalization ability. It ensures that the model utilizes all pathways instead of overfitting to the most dominant ones. To achieve this, it penalizes the model when outputs deviate from the expected results from different pathways. Pathway Regularization reduces the model's dependence on a single pathway and makes the learning process more robust and redundant. It is especially useful in techniques where multiple features may show the presence of a disease.

Conceptual Foundation

Multiple attention heads are employed in each layer in the transformer. Here, each head can be seen as a separate "pathway" through which the network learns different characteristics of the data. Similarly, different sequences of layers or blocks can form other pathways.

Mathematical Formulation of Pathway Regularization

We have proposed Pathway Regularization in equation 5. The Pathway Regularization can be expressed as:

$$L_{\text{pathway}} = \delta \sum_{\text{all paths}} (\text{expected output} - \text{actual output})^2 \tag{5}$$

Here: "all paths" means to different computational routes through the model (different heads, sequences of layers). The "expected output" is the target output for a given input that can represent the ground truth or a more abstract representation of what is expected from each pathway. The "actual output" is the output produced by the model for each pathway. The term δ is a tuning parameter that determines the weight of this regularization term relative to the main loss. The sum is taken over different computational pathways (different attention heads or different sequences of blocks).

Algorithm 2 Swin Transformer with Pathway Regularization

Require: Training dataset $D = \{(x_1, y_1), \ldots, (x_n, y_n)\}$, number of classes *num_classes*, regularization coefficient δ, number of epochs E, learning rate lr, batch size B
Ensure: Trained Swin Transformer model f with parameters θ

```
function PathwayRegularization(expected_output, actual_output, δ)
    pathway_loss ← δ · ∑(expected_output − actual_output)^2
    return pathway_loss
end function
function SwinTransformerModel(num_classes, embed_dim, num_heads)
    patch_embedding ← Initialize patch embedding layer
    swin_block ← Swin Transformer block with embed_dim and num_heads
    classifier ← Linear(embed_dim, num_classes)
    return Compound model of patch_embedding, swin_block, and classifier
end function
model ← SwinTransformerModel(num_classes, embed_dim=96, num_heads=3)
model.to(device)
optimizer ← Adam(model.parameters(), lr = lr)
criterion ← CrossEntropyLoss()
for epoch = 1 to epochs do
    logs ← model.fit(train_generator)
end for
```

Note: The pathway regularization is applied to the output obtained after attention and optimization adjustments.

Pathway Outputs

It computes the outputs from all defined pathways for each input. The transformer with multiple heads might involve isolating the output contribution of each head.

Expected vs. Actual Output

1. The "expected output" for each pathway is based on the overall target or its transformed version. It shows what each pathway should ideally produce.
2. The "actual output" directly comes from the model for each pathway.

Regularization Term Calculation

1. It computes the squared difference between the expected and actual outputs, for each pathway.
2. Finally, sum these squared differences across all pathways.

Integration with Loss Function

It adds the pathway regularization term to the main loss function of the model:

$$L = L_{\text{main}} + L_{\text{pathway}}$$

Here, L_{main} shows the standard loss term similar to cross-entropy for classification.

Expected Output Formulation:

For intermediate computational pathways, no explicit control signal exists, so the expected output is estimated using a statistical approximation over the mini-batch. especially, for a batch of size B, the expected output is computed as:

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$$\hat{y}_{\text{exp}} = \frac{1}{B}\sum_{i=1}^{B} y_i \tag{6}$$

here y_i denotes the model output for the i-th sample in the batch. This technique works as a self-consistency condition. This encourages individual pathway outputs to align with the collective representation of the batch. It also bypasses the need for explicit supervision at intermediate layers and provides a meaningful regularization signal.

5.3. Curriculum Learning

The Curriculum learning first starts working with local attention and slowly includes a more global context. It dynamically adjusts the softmax scaling factor in training epochs. A curriculum learning approach is useful when the model first learns from local attention only and then slowly shifts to more complex attention patterns, such as full self-attention. It introduces complexity in a controlled manner for Attention, which helps the model to gradually learn to identify and classify the leaf disease categories. This approach is useful to avoid overfitting with less complex or less varied training data.

Base Self-Attention Formula: The standard self-attention mechanism in transformers is as follows:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{7}$$

Modification with $\beta(t)$:

A scheduling parameter $\beta(t)$ that adjusts the scope of the attention mechanism over epochs. Initially, $\beta(t)$ makes the softmax function operate over a narrower range of values by scaling down the dot products. Hence, the attention mechanism is made "softer" or more local. The range of softmax grows as $\beta(t)$ increases, enabling a broader and more global attention distribution. This is governed by a scheduling parameter $\beta(t)$ that adjusts the scope of attention over epochs:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{\beta(t)\cdot QK^T}{\sqrt{d}}\right)V \tag{8}$$

Where: $\beta(t)$ is a function of training time (epoch number) t, and it adjusts the influence of the dot product QK^T over time.

Base Dot Product Calculation

It computes the matrix $S = QK^T$ and represents the raw alignment scores between queries and keys.

Scaling by Dimensionality and $\beta(t)$

The scores are scaled by $\sqrt{d_k}$ for stability. It introduces $\beta(t)$, the scores are further modified to:

$$S' = \frac{\beta(t)\cdot S}{\sqrt{d_k}}$$

Algorithm 3 Swin Transformer with Curriculum Learning

Require: Training dataset $D = \{(x_1, y_1), \ldots, (x_n, y_n)\}$, number of classes *num_classes*, number of epochs E, initial β value β_0, learning rate lr, batch size B
Ensure: Trained Swin Transformer model f with parameters θ

function BetaScheduler(epoch, max_epochs)
  return $\beta_0 + (1.0 - \beta_0) \cdot \frac{\text{epoch}}{\text{max_epochs}}$
end function
function SwinTransformerModel(num_classes, embed_dim, num_heads)
  patch_embedding ← Initialize patch embedding layer
  swin_block ← Swin Transformer block with *embed_dim* and *num_heads*
  classifier ← Linear(embed_dim, num_classes)
  return Compound model of patch_embedding, swin_block, and classifier
end function
function SwinTransformerBlock(x, β, embed_dim, num_heads)
  normalized_x ← LayerNorm(x)
  Q, K, V ← normalized_x, normalized_x, normalized_x
  scaled_qk ← $\beta \cdot \frac{QK^T}{\sqrt{d}}$ ▷ Curriculum-scaled attention mechanism
  attention_weights ← softmax(scaled_qk, dim=-1)
  attention_output ← *attention_weights* · V
  return attention_output
end function
model ← SwinTransformerModel(num_classes, embed_dim=96, num_heads=3)
model.to(device)
optimizer ← Adam(model.parameters(), lr = lr)
criterion ← CrossEntropyLoss()
for *epoch* = 1 to *epochs* **do**
  logs ← model.fit(train_generator)
end for

Note: This modified attention is passed to subsequent modules for regularization and optimization.

Softmax Application:

It uses the softmax function to S' and spreads the attention as $\beta(t)$ changes. Initially, if the values $\beta(t) < 1$, then the softmax will be more uniform (local focus). It is slowly shifting to sharper distributions as $\beta(t)$ increases. The main objective of varying $\beta(t)$ is to allow the model to focus on local features initially. It then gradually learns to incorporate more complex global patterns, which can be essential for contextual understanding. The $\beta(t)$ starts with a value less than 1 and gradually increases to 1 or more, depending on the desired learning progression. It can be defined in various ways depending on the desired curriculum speed and complexity:

- Linear: $\beta(t) = 1 + kt$ where k is a small constant.
- Exponential: $\beta(t) = \exp(kt)$ for faster increase.
- Step function: Increment $\beta(t)$ in steps, aligning with phases in the training process.

We represent a curriculum learning approach that dynamically adjusts the effect of the attention mechanism within a Swin Transformer model during training.

Model Architecture

The Swin Transformer has a feature extraction phase, which is followed by an attention mechanism. Here, the attention calculation is changed by a factor $\beta(t)$, which scales with the training epoch.

Attention Mechanism

The attention function is defined as:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{\beta(t)\cdot QK^T}{\sqrt{d_k}}\right)V \tag{9}$$

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Where: Q, K, V are the query, key, and value matrices, respectively. $\beta(t)$ is a scaling function of the epoch number t, adjusting the influence of QK^T over training time.

The concept of Curriculum Learning reasoning that humans learn more effectively when information is presented in order. The information starts with simple concepts and then moves to complex ones. It makes Transformer's training more efficient by initially giving simpler or less noisy instances in the training data, then the more complex ones. It helps to faster convergence and higher overall model performance, especially where random data presentation might initially dominate the model. The shifted window mechanism does not control the advancement of learning complexity during training. If the model is exposed to full attention in the initial epochs, the patterns can lead to unstable convergence. The curriculum learning in this research slowly increases attention complexity using a scaling parameter $\beta(t)$. This enables the model to learn stable low-complexity patterns in early stages and slowly adapt to more complex attention structures. Hence, the proposed approach complements the shifted window mechanism by introducing a training-level control over attention dynamics.

5.4. Adaptive Learning Rate Scheduling

The idea of making the learning rate adaptive based on attention patterns is to analyse the output of softmax's standard deviation across various layers. The standard deviation clearify that how the attention weights are distributed, where a low standard deviation means concentrated attention, and a high standard deviation displays dispersed attention. According to the attention patterns in the training, the Rate of Adaptive Learning is adjusted. In the training, the learning rate could decrease if the attention becomes too concentrated. This could also be the reason for finding other features. As per the model attention performance, the Adaptive Learning controls the learning rate, stabilizes training, and improves convergence. This adjustment can optimize the learning process, particularly when identifying the subtle differences between healthy and diseased leaf imagery.

Mathematical Formulation of Adaptive Learning Rate Scheduling:

The Adaptive Learning Rate Scheduling can be expressed as:

$$lr = lr_0 \exp\left(-\eta \sum \text{std}\left(\text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)\right)\right) \tag{10}$$

Here, lr_0 is the initial learning rate and η is a sensitivity parameter which defines how responsive the learning rate is to changes in the standard deviation of the attention weights. The std$(\cdot)$ calculates the standard deviation of the softmax outputs and shows how focused or spread out the attention is.

Attention Weight Calculation

Compute the dot products:

$$S = \frac{QK^T}{\sqrt{d_k}}$$

Algorithm 4 Swin Transformer with Adaptive Learning Rate

Require: Training dataset $D = \{(x_1, y_1), \ldots, (x_n, y_n)\}$, number of classes *num_classes*, number of epochs E, initial learning rate lr_0, sensitivity parameter η, batch size B
Ensure: Trained Swin Transformer model f with parameters θ

```
function AdaptiveLRScheduler(optimizer, attention_weights, η)
    std(curriculum-adjusted attention_weights) ← std(softmax(QK^T/√d_k))
    new_lr ← lr_0 · exp(−η · std_attention)
    for each param_group in optimizer.param_groups do
        param_group['lr'] ← new_lr
    end for
end function
function SwinTransformerModel(num_classes, embed_dim, num_heads)
    patch_embedding ← Initialize patch embedding layer
    swin_block ← Swin Transformer block with embed_dim and num_heads
    classifier ← Linear(embed_dim, num_classes)
    return Compound model of patch_embedding, swin_block, and classifier
end function
function SwinTransformerBlock(x)
    normalized_x ← LayerNorm(x)
    Q, K, V ← normalized_x, normalized_x, normalized_x
    attn_output, attn_weights ← MultiHeadAttention(Q, K, V)
    return attn_output, attn_weights
end function
model ← SwinTransformerModel(num_classes, embed_dim=96, num_heads=3)
model.to(device)
optimizer ← Adam(model.parameters(), lr = lr_0)
criterion ← CrossEntropyLoss()
for epoch = 1 to E do
    loss ← criterion(outputs, labels)
end for
return model
```

Apply softmax to get the attention weights:

$$A = \text{softmax}(S)$$

Standard Deviation Computation

Each head or each batch of data it compute the standard deviation of the entries in the attention matrix A. It sums these standard deviations across all heads or layers to get a total measure of attention spread.

Learning Rate Adjustment

The standard deviation is used to adjust the learning rate exponentially. Here, the negative sign in the exponent verifies that a higher std deviation leads to a smaller adjustment to lr_0. Whereas the lower std deviation results in a huge decrease in the learning rate. It involves adjusting the learning rate in training, where in the early training stages, a larger learning rate enables faster convergence, and then it is lowered to enable finer corrections. Adaptive Learning Rate Scheduling strategies help transformers in learning rate decay. The more complex approaches, such as learning rate schedulers, also help to adjust the rate based on performance parameters. The technique provides stable training dynamics and avoids local minima, which is necessary while considering the complexity of loss functions in Transformers.

The adaptive learning rate scheduling is applied only during the training phase. For each mini-batch training, the standard deviation of the attention weights is computed and used to dynamically update the optimizer learning rate. This adaptive mechanism is caused only by training-batch statistics and does not include any information from validation or test data. During validation and testing, the model parameters remain fixed, and the learning rate scheduler is disabled. As no parameter updates

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Table 4: Summary of established foundations, attention-guided adaptations, and contribution scope of the ST-APCAR framework.

Component	Specified Basis	Adaptation in ST-APCAR	Novelty Scope
Swin Transformer Backbone	Hierarchical shifted-window transformer architecture	Used as the base feature extractor for leaf disease classification	Adopted backbone, not claimed as a new transformer architecture
Attention Norm Regularization	Regularization principles and attention-weight control	Penalizes deviation of attention distributions from a reference uniform distribution: $L_{\text{ANR}} = \lambda \sum_i \|A_i - U\|_2^2$	Mathematical adaptation applied directly to Swin attention distributions
Pathway Regularization	Consistency-based and regularization-based learning	Introduces a pathway-level penalty to reduce dependence on dominant attention heads or computational routes	Adapted regularization objective for attention-pathway stability
Curriculum Learning	Progressive learning from simpler to harder representations	Introduces epoch-dependent attention scaling: $\text{softmax}\left(\frac{\beta(t)QK^T}{\sqrt{d_k}}\right)$	Training-level adaptation for controlling attention complexity over epochs
Adaptive Learning Rate Scheduling	Learning-rate scheduling and adaptive optimization	Updates learning rate using attention-dispersion statistics: $lr = lr_0 \exp(-\eta \sum \text{std}(A))$	Attention-conditioned learning-rate adaptation rather than fixed epoch-based decay
ST-APCAR Framework	Integration of transformer training strategies	Combines attention regularization, pathway consistency, curriculum-scaled attention, and attention-driven learning-rate adaptation	Framework-level contribution through coordinated attention-guided training dynamics

Table 5: Computational efficiency comparison between vanilla Swin Transformer and ST-APCAR.

Model	Params (M)	FLOPs (G)	Inference Time (s)	Train Step Time (s)
Vanilla Swin Transformer	28.27	4.37	0.0154	–
ST-APCAR	32.24	4.60	0.0168	0.1005

appear during inference, the learning rate has no operational role in the testing phase. Based on the validation performance, the Model selection is performed, and the test set is used only once for final evaluation. This protocol ensures rigid separation between training and evaluation processes, prevents data leakage, and maintains fairness. Further, all models are trained and evaluated under identical conditions, which include the same data splits, optimizer configurations, and training epochs. The only difference lies in the proposed attention-driven adaptive learning rate strategy, ensuring a fair and controlled comparison.

Optimization and Stability Analysis of ST-APCAR

The components of ST-APCAR are designed as attention-guided training constraints instead of closed-form optimization. As the Swin Transformer is a deep non-convex model, providing a strict global convergence proof is difficult. The main objective is to control the internal attention behavior of the model to improve the stability and regularity of the training process. The Attention Norm Regularization works on softmax-normalized attention distributions where every attention value is specified within $[0, 1]$. Hence, the regularization term remains bounded and provides a stable auxiliary penalty. Furthermore, the curriculum learning parameter $\beta(t)$ is presented as a bounded epoch-dependent factor. This factor slowly changes the sharpness of attention logits. In the early training stage, it avoids abrupt exposure to highly complex attention patterns and provides smoother optimization. The attention weights are distributed as an internal training signal by the adaptive learning rate. The learning rate is constrained within a predefined range to maintain stability, so that it can prevent excessively large or vanishing updates. The Pathway regularization reduces over-dependence on dominant attention routes, and it is also used as a self-consistency constraint across computational pathways. Thus, the computational cost stays the same as the original window-based self-attention operation. The additional cost due to computing attention statistics, pathway consistency, and auxiliary regularization losses. The regularization and learning-rate adaptation are disabled during testing. It has shown a moderate training-time overhead and no additional inference-time cost when

6. Results Analysis& Discussion

The result analysis section discussed the Experimental Setup, Computational Efficiency Analysis, Data Splitting Strategy, Hyperparameter Settings, Module usage, and the evaluation metrics such as Accuracy, Loss, Precision, Recall, F1, and Log Loss on AG and PD datasets.

Experimental Setup

The experiment setup is based on the NVIDIA DGX-1 V100 system, which has 8× NVIDIA Tesla V100 GPUs and is equipped with 16GB HBM2 memory and 128GB GPU memory. The system is also quipped with dual Intel Xeon E5–2698 v4 CPUs, 512GB of RAM, and 7.68TB of SSD storage. Popular libraries such as TensorFlow, Keras, Matplotlib, and OpenCV are used with the Python language. To maintain reproducibility, we fixed the random seed for dataset splitting, model initialization, and during model training. We also used Stratified sampling during the model training, validation, and testing to maintain the class distributions. We employed data augmentation on train set, and Hyperparameter tuning, adaptive learning rate adjustment, and model selection were used in the validation set only. We also employed early stopping on the validation and test set used only for final analysis. Hence, preventing data leakage and ensures an unbiased assessment of generalization performance.

Computational Efficiency Analysis

The baseline Swin Transformer and the proposed ST-APCAR framework computational complexity was evaluated based on parameter count, FLOPs, inference time, and training step time. Through training and optimization level components, Attention Norm Regularization, Pathway Regularization, Curriculum Learning, and Adaptive Learning Rate Scheduling, the

framework enhances Swin Transformer. The framework does not present an architectural compression mechanism such as pruning, quantization, knowledge distillation, or parameter reduction. Hence, the complexity of the model is predicted to be relative to the baseline Swin Transformer. The small increase in parameter count and FLOPs in ST-APCAR is due to the implementation-specific classifier and auxiliary training-support components that are used to enable the attention-guided training framework, as shown in Table 5. This overhead is due to the auxiliary regularization and optimization components used during training. Whereas the inference time stays close to the baseline model. This shows that the proposed framework maintains the computational characteristics of the Swin Transformer while improving classification performance. The components are integrated with the Swin Transformer backbone without replacing, compressing, or redesigning the original backbone. Therefore, the increase from 28.27M to 32.24M parameters shows an overhead, while the main feature extraction architecture remains Swin Transformer-based.

Hyperparameter Settings and Module Usage

The proposed framework, based on four hyperparameters, is: Attention Norm Regularization (λ), Pathway Regularization (δ), Curriculum Learning (β), and Adaptive Learning Rate Scheduling (η). To achieve a balance between convergence stability and classification performance, these hyperparameters were specified using validation-based tuning. The four strategies were evaluated independently and then combined in the final framework to examine the effectiveness. The $\delta = 0.01$was selected for Pathway Regularization because it delivered the most favorable balance between validation loss and predictive performance. Other hyperparameters λ, β, and η, were selected according to their validation-set behavior and their contribution to stable learning.

Data splitting strategy

To split the dataset, we have used (70%) training, (15%) validation, and (15%) testing based on stratified sampling on class labels to avoid data leakage. This also makes sure the class distribution is maintained across all subsets. To avoid the overlapping between training, validation, and testing data, the splitting is performed before data augmentation. Similarly, to get an unbiased evaluation of the model and its generalization capability, all preprocessing and augmentation techniques were applied only to the training set.

6.1. *Dataset-wise Performance Analysis*

This section analyzes the Accuracy, Loss, Precision, F1-Score, MCC, and log loss as shown in Fig. 3 and Fig.4 of the proposed techniques for AG and PD datasets based on equations 7, 5, 8, and 10.

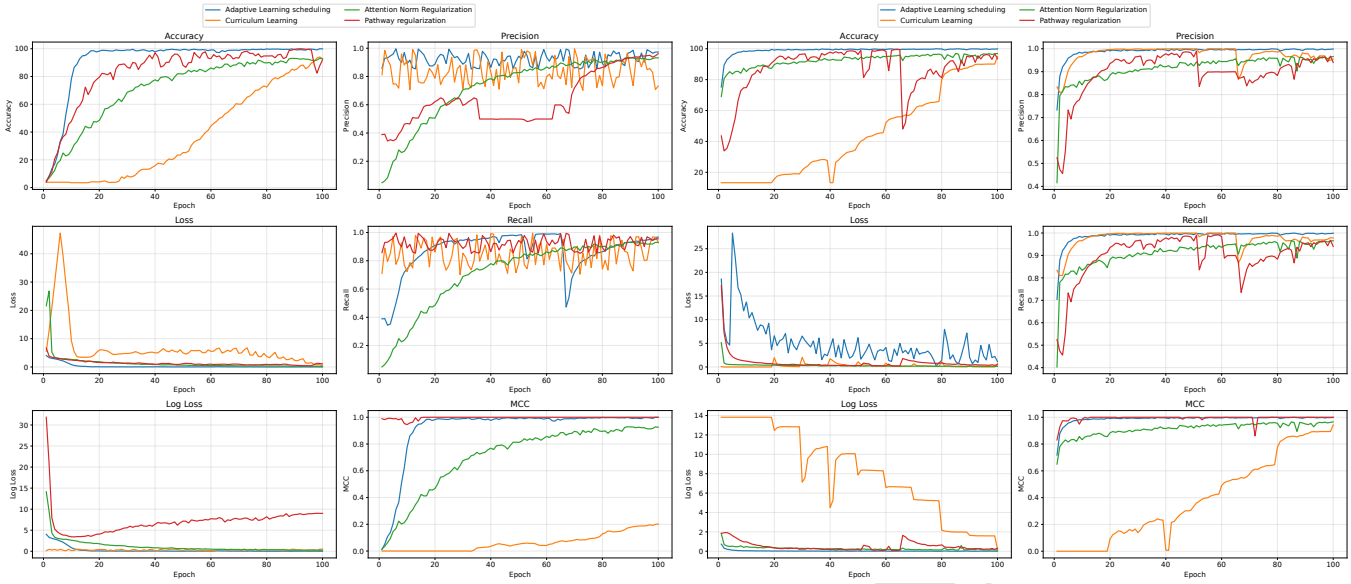
6.1.1. *AG dataset*

Curriculum learning consistently is low, which may be misrecorded or represented quite differently. Both Pathway Regularization and Attention Norm Regularization follow typical machine learning behavior, where models start with low performance and gradually improve as they learn from more data. Adaptive learning demonstrates a steady, progressive improvement, particularly when learning needs to be regulated and stabilized. The importance of Adaptive Learning Rate Scheduling begins at the 10th epoch, where accuracy reaches 78%, and by the 100th epoch, it approaches near-perfect accuracy of about 99.9%. This result demonstrates the advantage of learning and adaptation over time. Nevertheless, Curriculum Learning has a distinctive accuracy pattern, possibly indicating a different scale, with values ranging from 3.5 to 92.76. Attention Norm Regularization with about 26.6% accuracy at the 10th epoch improves to around 92.7% by the 100th epoch. This shows that attention norms improve the model's learning capacity. Pathway Regularization with 47.7% accuracy at the 10th epoch reaches 97.9%-91.9% in the 100th epoch. This approach jumps very quickly, but then decreases in a pattern afterwards.

Adaptive Learning Rate Scheduling begins with a loss of 0.84 at the 10th epoch, finally stabilizing at low values of 0.0056 at the 100th epoch. This shows that learning is highly effective, and errors are minimized. Curriculum Learning starts with a loss of 9.27 at the 10th epoch, slowly dropping to a value of 0.38 by the 100th epoch. After significant fluctuations, the pattern shows a significant reduction, showing good learning. Attention Norm Regularization starting from 2.65 at the 10th epoch shows a smooth and uniform reduction in loss to 0.27 by the 100th epoch. The gradual decrease shows that regularization techniques improve the model's generalization capacity and reduce overfitting. Pathways Regularization begins with a loss of 2.49 at the 10th epoch, then rises slightly again to 1.10 by the 100th epoch. This change shows that the model is continuously learning throughout the epoch, but there are examples where learning is less effective.

The adaptive Learning shows the strongest precision at 0.9732 with a high mean precision of about 0.9242 across epochs. Whereas the Pathway Regularization reaches about 0.9608 at the final epoch, showing the lower average performance because of its weaker early stages. The Attention Norm Regularization starts at 0.0480 and ultimately reaches 0.9319. Similarly, the Curriculum Learning has strong precision at the beginning, with a final precision of 0.7324, which is lower than the other techniques. It shows how behavior varies over the training in comparison to Adaptive Learning. Finally, on the AG dataset, the Adaptive Learning provides the most appropriate balance of high precision and stable convergence. Whereas the Attention Norm Regularization shows the most affected recovery from a weak initial state.

For the Recall the Pathway Regularization has shown the mean recall of 0.9226, ending with a strong final recall near 0.9558 at the 100th epoch. Adaptive Learning with a recall of about 0.9678 shows more significant variation over epochs in comparison to the Pathway Regularization. The Curriculum Learning shows good but comparatively lower recall performance at the final epoch, with a 0.9340 recall value. Whereas the Attention Norm Regularization shows great progress, starting with a weak 0.0482 value to 0.9288 at the final epoch. The slow but constant progress across the epoch shows that the



(a) AG dataset

(b) PD dataset

Figure 3: Accuracy, Precision, Loss, Recall, Log Loss, MCC metrics for Proposed techniques on (A) AG dataset and (B) PD dataset.

model is slowly becoming capable of classifying classes accurately. At the end, the results show that the Pathway Regularization and Adaptive Learning are the strongest methods. The Adaptive Learning shows the highest final recall, and Pathway Regularization offers the most consistent recall across training.

The F1-score, by penalizing both false positives and false negatives, balances precision and recall. It is important for datasets that have uneven class distributions. The Adaptive Learning with highest mean F1-score of 0.9369 and reached 0.9899 at the 100th epoch. The Pathway Regularization shows a mean F1-score of 0.9040 and 0.9899 in the final performance because of less consistent early behaviour. Whereas the Curriculum Learning shows a moderate but balanced between precision and recall, reaching 0.8726 at the final epoch. The Attention Norm Regularization with major performance recovery over the epochs improves from 0.0359 to 0.9292. Hence, the Adaptive Learning shows the most suitable F1-score behaviour, and Pathway Regularization is a strong competing technique with outstanding final convergence.

Adaptive Learning Rate Scheduling changes the learning rate dynamically based on the loss gradient and training epoch. It stabilizes and speeds up the training process by optimizing the learning rate. Dynamic adjustment saves the model from going into local minima close to the global minimum. This gives better convergence and better classification results. Curriculum Learning with an MCC of 0.2019 presents the model to complex data samples. The low MCC means that the model was unable to manage the complexity caused by the curriculum latter failed to fit the data well. Attention Norm Regularization in the loss function uses penalties and adjusts the size of parameters to prevent overfitting of the model. The high MCC value shows that the model achieved robust features for predicting and generalizing accurately. Regularization also helps to control the complexity of models, reduce variance, and control bias. Pathways Regularization with an MCC of 1.0 is considered the best learning outcome obtained.

Dynamic adaptive learning rates start from 0.84 and end up at 0.01 at epoch 100. This will allow the model to be precise about finding the minimum point in the loss function. This is seen to have an extremely low log loss rate, along with predictions that match the actual classes. The model was able to adapt and contribute to minimizing the log loss. Curriculum Learning with 0.49 at epoch 100 increased model complexity by revealing the model to different patterns that would not overwhelm the model initially. Attention Norm Regularization with 0.26 at epoch 100 minimizes large weight matrices. It decreases the possibility of overfitting and makes the model learn more natural patterns. Pathway Regularization uses informative subsets to ensure the model focuses on the most informative features. This helped reduce the log loss and made accurate classifications possible.

6.1.2. PD dataset

Adaptive Learning rate Scheduling starts with 98.19% accuracy on the 10th epoch with few deviations, reaching 99.89% by the 100th epoch. Attention Norm Regularization starts at 85.21% in the 10th epoch and increases to 96.92% by the 100th epoch. This shows that attention-based regularization supports model learning. Curriculum Learning starts a bit randomly with 13.28% at the 10th epoch, grows upwards to 94.93% at the 100th epoch, hence this shows a good learning curve that is supported by curriculum-based epochs. Pathway Regularization starts at 74.82% at the 10th epoch, goes to 98.33% at the 60th epoch, then drops to 93.52% at the 100th epoch. This suggests that the heterogeneity of learning is due to the adjustments to model parameters or their adaptation techniques.

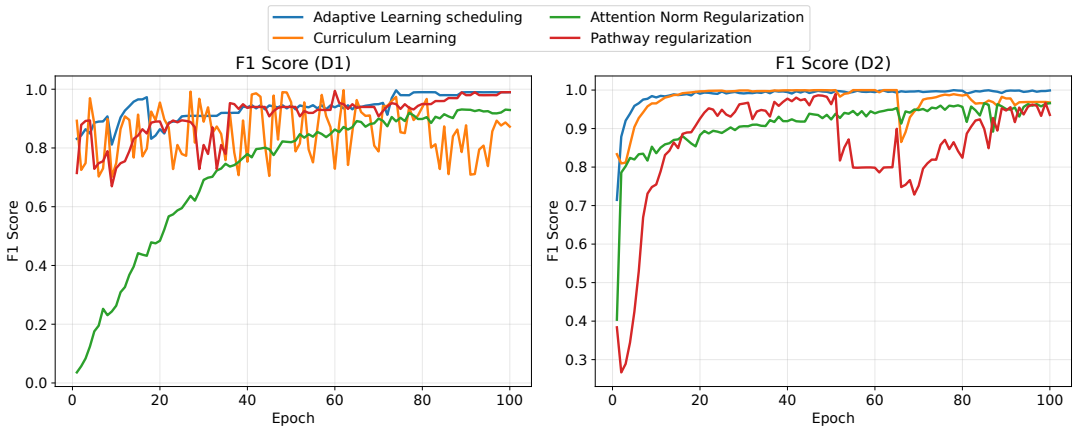


Figure 4: F1-Score for Proposed techniques on (A) AG dataset and (B) PD dataset

The Attention Norm Regularization starts with a loss of 0.4553 and at the 100th epoch declines to 0.1261. This shows how the how strong and consistent the regularization methods are when the training progresses. The Adaptive Learning Scheduling with a loss of 13.7450 at the 10th epoch and drops to 1.1317 at the end of the training. These declines show that the model effectively optimizes and minimizes the errors over time. Whereas the Curriculum Learning starts with a loss of 0.000 and shows a good performance on a subset. As training progresses, the loss decreases to 0.7404 values by the 100th epoch. This trend explains that the model stabilizes due to the tasks that were too difficult. The Pathway Regularization with 1.2299 loss decreases at the 40th epoch and further goes to 0.4618 at the end of training. This shows efficient learning combined with facts when the model struggled with certain data before regaining efficiency.

The Adaptive Learning reaches the highest precision of 0.9989 on the PD dataset with the highest average precision across epochs as well. Whereas Curriculum Learning performs quite well with a mean precision of 0.9752. Although its final precision is slightly lower at around 0.9547. The Attention Norm Regularization increases from 0.4158 to 0.9660, and Pathway Regularization increases from about 0.5250 to 0.9402. Altogether, the PD dataset shows stronger precision performance, with Adaptive Learning emerging as the most reliable and effective technique.

The Adaptive Learning achieved the highest final recall of 0.9989 and the highest average recall of 0.9881 on the PD dataset. Curriculum Learning scores mean recall and final recall of around 0.9762 and 0.9778, respectively. Attention Norm Regularization grows from about 0.4029 to 0.9657, and by the end of training, the Pathway Regularization gains about 0.9402. Finally, the PD dataset shows perfect recall performance, and Adaptive Learning also proves the most effective technique at high recall levels.

The Adaptive Learning reaches the highest final and mean F1-score on the PD dataset, both at around 0.9989 and 0.9882, respectively. Curriculum Learning observes with a mean F1-score of about 0.9760 and a final value near 0.9678. Attention Norm Regularization increases from around 0.4037 to 0.9658, while Pathway Regularization rises from approximately. 0.3839 to 0.9355. These findings suggest that the PD dataset allows stronger overall F1-score performance than the AG dataset, and Adaptive Learning is clearly the most effective technique for maintaining the best precision-recall balance.

Attention Norm Regularization introduced a new way of training through imposing a penalty for model complexity using MCC 0.9626. Attention Norm Regularization is interested in learning the most significant features using the method of norm-based regularization. The high MCC value (0.9626) indicates that the model retained its robustness despite using complex datasets. Pathway Regularization is indicated as having very high performance of feature selection and prediction with a perfect MCC of 1.00. Through minimizing the effect of irrelevant data, Pathway Regularization was able to emphasize the significance of important features, leading to a perfect prediction outcome.

The adaptive learning rate scheduling helped the learning rate to be adjusted and decreased to 0.01 throughout the training period, when an optimal solution had been reached by the model. The log loss is extremely low, meaning that predictions done by the model were very confident and accurate. At the 100th epoch, attention norm regularization of 0.12 minimized the problem of overfitting and increased the generalization capacity by focusing on the most important features. The extremely low log loss underscores the model's strong predictive performance. Pathway regularization of 0.311 at the 100th epoch emphasizes the importance of selecting only those features that help increase prediction accuracy.

7. Effect of the Proposed Training Strategy

This section discusses the overall comparative performance on the AG and PD dataset for metrics such as Accuracy, Precision, Recall, F1-Score, Log loss, MCC, and class-wise analysis.

Adaptive Learning and Pathway Regularization show the strongest convergence behaviour on the AG dataset. Adaptive Learning in Fig. 3 shows a clear rise in accuracy from a very low initial value to nearly perfect. Whereas its loss

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Table 6: Metrics for Different Learning Strategies at Every 10th Epoch on AG dataset

Epoch	Adaptive Learning Scheduling				Curriculum Learning				Attention Norm Regularization				Pathway Regularization			
	Accuracy	Loss	MCC	Log Loss	Accuracy	Loss	MCC	Log Loss	Accuracy	Loss	MCC	Log Loss	Accuracy	Loss	MCC	Log Loss
10	0.78	0.84	0.77	0.84	0.03	9.23	0.00	0.46	0.26	2.64	0.24	2.60	0.47	2.49	0.95	0.343
20	0.98	0.07	0.98	0.07	0.04	4.10	0.00	0.37	0.48	1.82	0.47	1.78	0.79	1.64	0.95	0.406
30	0.99	0.03	0.99	0.03	0.09	6.63	0.00	0.25	0.69	1.15	0.68	1.14	0.87	1.41	0.94	0.558
40	0.98	0.07	0.98	0.07	0.15	5.00	0.04	0.38	0.78	0.88	0.77	0.82	0.97	0.92	0.92	0.641
50	1.00	0.02	1.00	0.02	0.23	6.08	0.05	0.25	0.82	0.63	0.81	0.64	0.88	1.19	0.94	0.716
60	0.99	0.04	0.99	0.04	0.44	6.00	0.04	0.12	0.86	0.49	0.86	0.49	0.93	0.99	0.91	0.773
70	0.99	0.04	0.99	0.04	0.60	6.62	0.08	0.19	0.90	0.40	0.90	0.38	0.93	0.99	0.93	0.761
80	1.00	0.00	0.99	0.00	0.73	4.73	0.613	0.31	0.90	0.36	0.89	0.34	0.95	0.92	0.91	0.821
90	0.99	0.02	0.99	0.02	0.85	3.15	0.518	0.46	0.93	0.26	0.93	0.27	0.93	0.61	0.93	0.871
100	0.99	0.01	0.98	0.01	0.92	0.38	0.520	0.49	0.92	0.27	0.93	0.26	0.91	1.10	0.92	0.65

Table 7: Metrics for Different Learning Strategies at Every 10th Epoch on PD dataset

Epoch	Adaptive Learning Scheduling				Curriculum Learning				Attention Norm Regularization				Pathway Regularization			
	Accuracy	Loss	MCC	Log Loss	Accuracy	Loss	MCC	Log Loss	Accuracy	Loss	MCC	Log Loss	Accuracy	Loss	MCC	Log Loss
10	0.98	0.1374	0.98	0.05	0.1328	0.00	0.00	0.1382	0.85	0.46	0.83	0.48	0.75	1.22	0.98	0.97
20	0.99	0.660	0.99	0.03	0.1566	1.97	0.09	0.1247	0.90	0.40	0.88	0.34	0.92	0.59	0.95	0.37
30	0.99	0.653	0.99	0.03	0.2141	2.03	0.14	0.712	0.91	0.30	0.90	0.29	0.96	0.43	1.00	0.22
40	0.99	0.409	0.99	0.02	0.1330	1.74	0.01	0.451	0.93	0.25	0.92	0.26	0.98	0.33	1.00	0.16
50	0.99	0.619	0.99	0.02	0.3781	1.09	0.33	0.786	0.93	0.24	0.92	0.25	0.96	0.33	0.99	0.19
60	0.98	0.411	0.99	0.02	0.5215	0.80	0.49	0.658	0.95	0.19	0.94	0.19	0.98	0.24	0.99	0.12
70	0.99	0.173	0.99	0.01	0.5978	0.71	0.58	0.534	0.95	0.17	0.95	0.17	0.73	1.25	0.80	1.02
80	0.99	0.138	0.98	0.01	0.7993	1.12	0.78	0.217	0.96	0.15	0.95	0.14	0.81	0.84	0.83	0.66
90	0.99	0.202	0.98	0.01	0.8838	0.27	0.87	0.164	0.96	0.16	0.95	0.17	0.95	0.43	0.94	0.26
100	0.99	0.113	0.99	0.01	0.9493	0.74	0.94	0.016	0.96	0.13	0.97	0.12	0.93	0.46	0.93	0.31

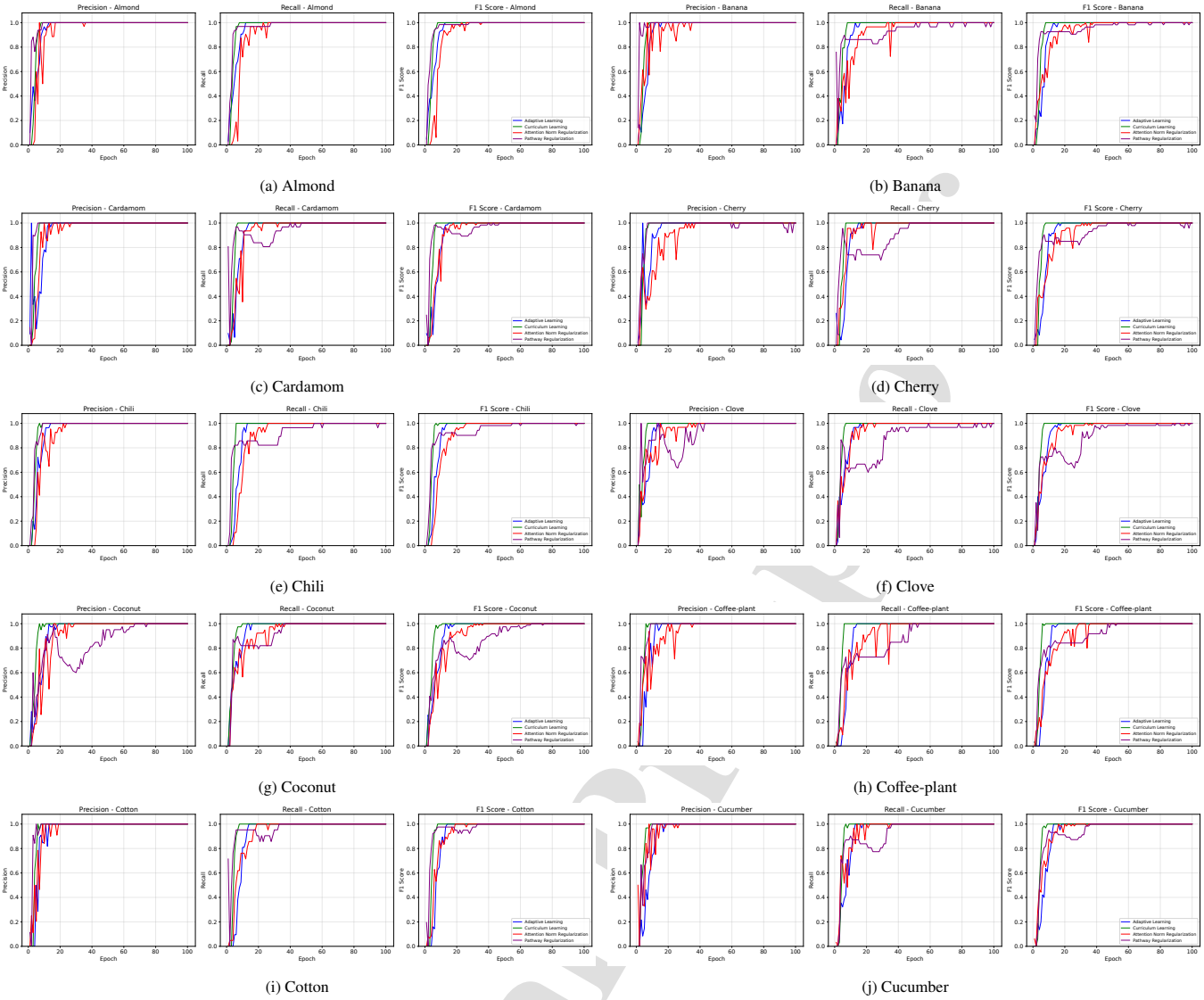
decreases sharply, showing effective optimization and stable learning. The Pathway Regularization also achieves superior accuracy with a significant decline in loss, showing a robust behaviour throughout training. The Attention Norm Regularization begins with weak performance but gradually improves over time, showing that the method understands effective feature representations. As training progresses, the Curriculum Learning improves but remains lower than Adaptive Learning and Pathway Regularization on the AG dataset. Attention Norm regularization, and pathway regularization are also highly effective, with clear gains in accuracy and the loss of accuracy decreasing over epochs. Curriculum Learning achieves competitive accuracy on the PD dataset, although its loss behaviour is less consistent than Adaptive Learning. Altogether, Adaptive Learning provides the best balance between maximum accuracy and minimal loss on both datasets, while Pathway Regularization performs well on the AG dataset. All techniques show better overall behaviour for the PD dataset than in the AG dataset.

Adaptive Learning with decreasing log loss and increasing MCC to a high level on the AG dataset during training shows(Fig 3) the most fitting performance. This shows that the model not only improves accuracy but also creates more reliable class assignments. Adaptive Learning shows low log loss and the highest MCC among the techniques for the PD dataset. This proves the strong predictive confidence and balanced classification ability. Whereas the Pathway Regularization shows a strong MCC score, although its log loss remains higher than that of Adaptive Learning. But the Attention Norm Regularization shows significant improvements in both metrics, with the reduction of log loss and the increase of the MCC. Similarly, the Curriculum Learning method has demonstrated poor MCC results with the AG dataset, where class predictions were not as balanced as expected. Curriculum Learning shows improvement on the PD dataset compared with the AG dataset, but it remains less effective than Adaptive Learning in terms of both confidence calibration and balanced classification. Therefore, the combined log loss and MCC analysis further sustains that Adaptive Learning is the most reliable technique across both datasets.

Analysis of precision, recall, and F1-score across (Fig 3, 4) both datasets shows that Adaptive Learning is the most effective technique among all. Adaptive Learning performs well in precision and F1-score, while Pathway Regularization appears as a strong alternative, as it shows high recall and F1-score for the AG dataset. Whereas the Attention Norm Regularization initially shows very weak performance and strong final results on the AG dataset. Its average performance stays lower because of the poor initial epochs. Similarly, the Curriculum Learning shows useful gains, but does not match the average performance of Adaptive Learning on the AG dataset. All techniques perform better on the PD dataset than the AG dataset, as the PD dataset is more favourable for stable training and class discrimination (Table 6-7.)

The Adaptive Learning achieves high precision, recall, and F1-score by the final epoch, showing excellent performance among all. Whereas the Curriculum Learning is the second-best method, in terms of recall and F1-score. Similarly, Attention Norm Regularization becomes highly competitive after adequate training, and Pathway Regularization results remain below those of Adaptive Learning and Curriculum Learning. Therefore, based on the combined evidence from precision, recall, and F1-score across both datasets, Adaptive Learning is considered the best-performing technique overall. It is the strongest average behaviour across epochs, and has the highest

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(a) Almond (b) Banana

(c) Cardamom (d) Cherry

(e) Chili (f) Clove

(g) Coconut (h) Coffee-plant

(i) Cotton (j) Cucumber

Figure 5: Class-wise Precision, Recall, and F1-score trends across 100 epochs for Attention Norm Regularization, Pathway Regularization, Curriculum Learning, and Adaptive Learning Rate Scheduling on the AG dataset (Class 1–10).

final classification quality. Pathway Regularization is useful on the AG dataset, especially in recall, while Curriculum Learning is more competitive on the PD dataset. Attention Norm Regularization is a great late-stage improvement, but it is less stable during the early training period.

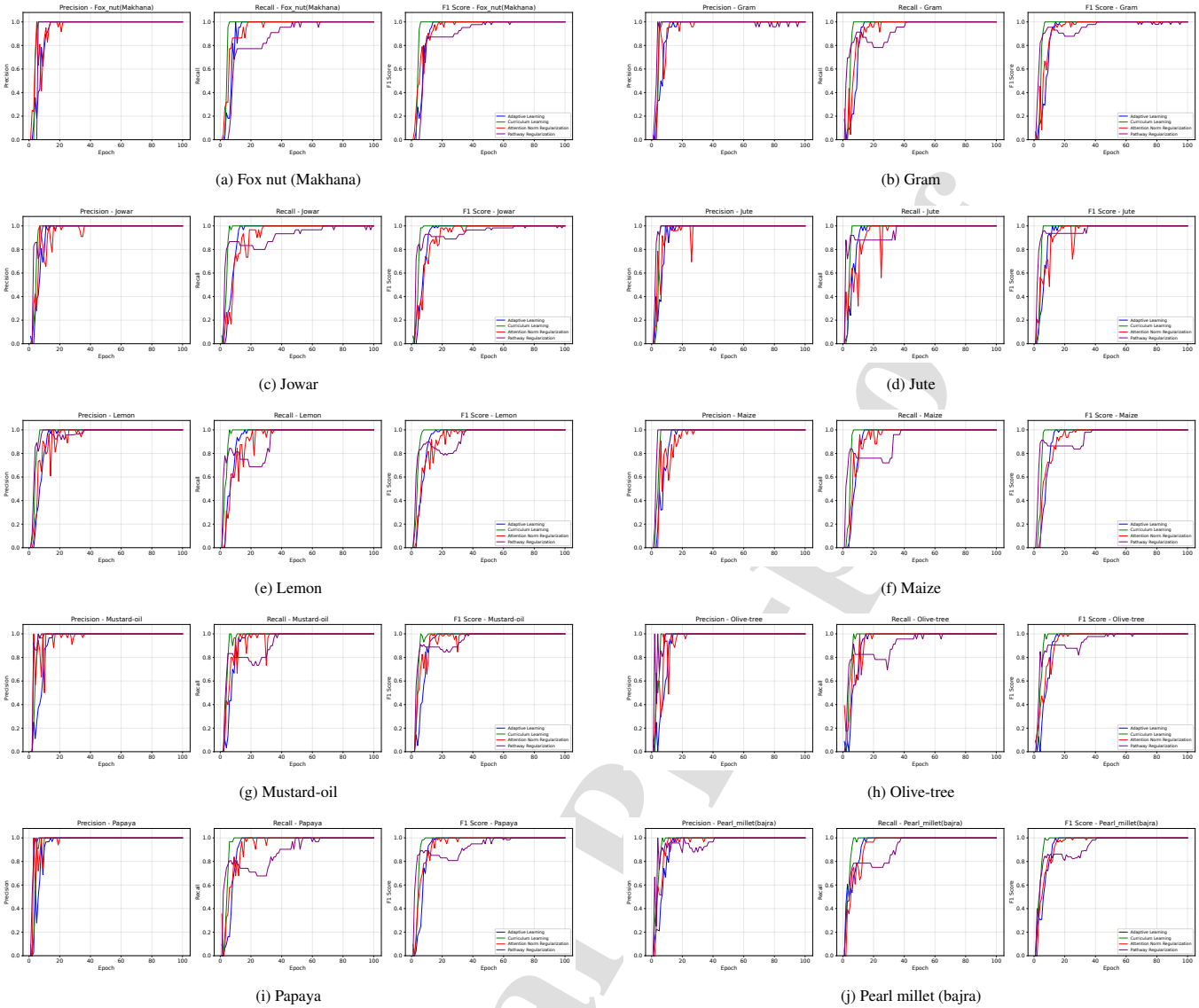
7.1. Class-wise Precision, Recall, and F1-score analysis

In this section, we have discussed the class-wise performance of Precision, Recall, and F1-score on the AG and PD dataset.

7.1.1. AG Dataset

The class-wise results for the AG dataset, as summarized in Table 9 and Fig. 5, 6, and 7 show that all four techniques gained high final performance across all classes. Unlike the PD dataset, where the gap between methods was more prominent, the AG dataset shows that all approaches are able to learn highly discriminative class representations by the end of training.

All methods yielded values very close to 1.0, for nearly all classes. The Curriculum Learning and Pathway Regularization, with several classes reaching 0.9999, showed consistently high final precision values. Curriculum Learning achieved near-perfect precision and shows reliable positive predictions for classes such as Banana, Cherry, Chili, Cotton, Cucumber, Jute, Olive-tree, Pearl millet (bajra), and Wheat. Whereas the Pathway Regularization also showed strong precision, especially for classes such as Almond, Banana, Cherry, Cotton, Jowar, Jute, and Sugarcane. In contrast, Adaptive Learning remained strong overall but had slightly lower precision for a few classes, such as Almond (0.9960), Pineapple (0.9960), and Mustard oil (0.9967). Attention Norm Regularization also shows good scores, but Almond, Pineapple, and Tea classes showed slightly lower precision. The Precision results confirm that Curriculum Learning and Pathway Regularization show a slight edge in fi-



(a) Fox nut (Makhana) (b) Gram

(c) Jowar (d) Jute

(e) Lemon (f) Maize

(g) Mustard-oil (h) Olive-tree

(i) Papaya (j) Pearl millet (bajra)

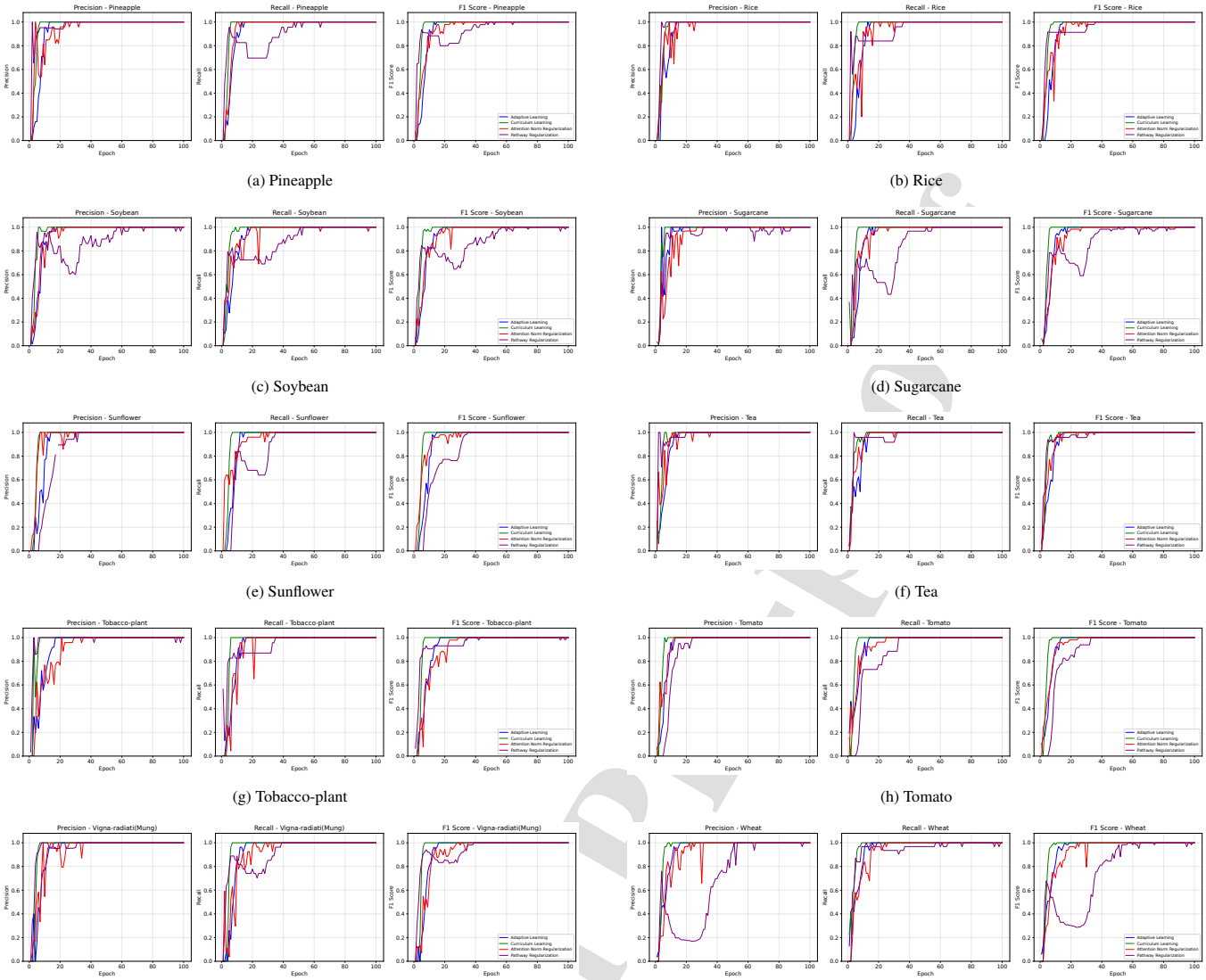
Figure 6: Class-wise Precision, Recall, and F1-score trends across 100 epochs for Attention Norm Regularization, Pathway Regularization, Curriculum Learning, and Adaptive Learning Rate Scheduling on the AG dataset (Class 11–20).

nal class-wise discrimination.

A similar progress is followed for Recall, where all methods achieved high final values on the AG dataset. Where Pathway Regularization shows approx 0.9999 for many classes, which include Almond, Cherry, Coconut, Cotton, Jowar, Jute, Mustard oil, Papaya, and Vigna radiati (Mung). The Curriculum Learning method also showed 0.9999 stable recall behavior with Banana, Cherry, Chili, Cucumber, Jute, Lemon, Olive-tree, Papaya, Pearl millet (bajra), Sugarcane, and Wheat classes.

Similarly, the Attention Norm Regularization remained very competitive, often exceeding 0.998 and reaching 0.9999 for several classes, including Cardamom, Chili, Coconut, Papaya, Soybean, and Vigna radiati (Mung). In contrast, Adaptive Learning achieves excellent recall across all classes, but was marginally lower for a few cases, such as Almond (0.9961), Fox nut (Makhana) (0.9961), Pineapple (0.9961), and Mustard oil (0.9968). These differences are small, but Pathway Regularization and Curriculum Learning provided slightly stronger final class coverage on the AG dataset.

The F1-score balances Precision and Recall, and the uniformly high values show that all techniques achieved a strong trade-off between false positives and false negatives. The Curriculum Learning method appeared as one of the strongest, with many classes approaching 0.9999, including Banana, Chili, Cucumber, Jute, Olive-tree, Pearl millet (bajra), and Wheat. Similarly, Pathway Regularization also carried excellent F1-scores of 0.9990 for nearly all classes, such as Cherry, Cotton, Jowar, Jute, Sugarcane, Vigna radiati (Mung), and others. Whereas the Attention Norm Regularization shows a strong F1-score for several classes, such as Cardamom, Papaya, Soybean, and Pearl millet (bajra). Whereas Adaptive Learning remained highly ac-



(a) Pineapple (b) Rice

(c) Soybean (d) Sugarcane

(e) Sunflower (f) Tea

(g) Tobacco-plant (h) Tomato

(i) Vigna radiati (Mung) (j) Wheat

Figure 7: Class-wise Precision, Recall, and F1-score trends across 100 epochs for Attention Norm Regularization, Pathway Regularization, Curriculum Learning, and Adaptive Learning Rate Scheduling on the AG dataset (Class 21–30).

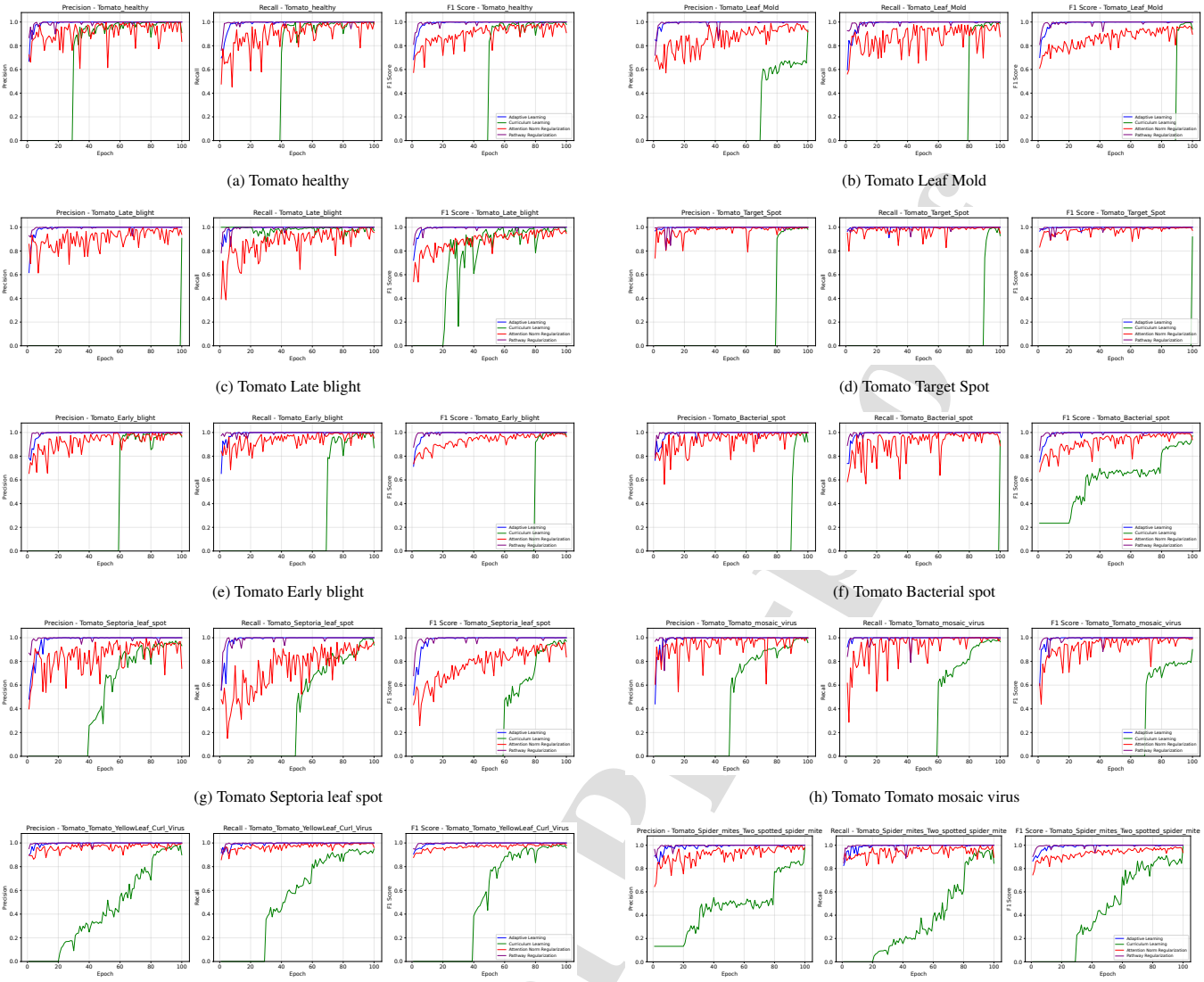
curate with a lower F1-score for certain classes, such as Almond (0.9960), Pineapple (0.9960), and Mustard oil (0.9967). Altogether, from a class-wise analysis, the AG dataset appears comparatively easier than the PD dataset. The techniques achieved near-perfect results on almost every category except a few classes, which showed slightly more variation across methods. For example, the Almond, Pineapple, Mustard oil, Fox nut (Makhana), and Tea categories had significantly low values on one or more approaches, which shows that they needed better feature differentiation. In contrast, the Cherry, Banana, Cucumber, Jute, Sugarcane, and Wheat groups scored very high across all approaches, showing that they could be clearly separated using the AG approach.

7.1.2. PD dataset

The PD dataset results reveal clear differences among the four training strategies in terms of final class-wise performance, consistency across categories, and overall robustness. As shown in, Fig. 8 and Table 8.

The Pathway Regularization shows the best results, as all classes are in the interval 0.9997-0.9999. The Adaptive Learning presented high precision stability for all classes and gave values close to 0.9990 for all classes. Whereas Attention Norm Regularization produced high values of Precision in Bacterial spot, Yellow Leaf Curl Virus, Mosaic virus, and Target Spot classes. This regularization performs worse with the more difficult classes, like Septoria leaf spot (0.7417), healthy (0.8363), and Leaf Mold (0.9178). Similarly, Curriculum Learning remained insufficient, with precision ranging between 0.8999 and 0.9826, and shows inferior performance with lower precision

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(a) Tomato healthy

(b) Tomato Leaf Mold

(c) Tomato Late blight

(d) Tomato Target Spot

(e) Tomato Early blight

(f) Tomato Bacterial spot

(g) Tomato Septoria leaf spot

(h) Tomato Tomato mosaic virus

(i) Tomato Tomato YellowLeaf Curl Virus

(j) Tomato Spider mites Two spotted spider mite

Figure 8: Class-wise Precision, Recall, and F1-score trends across 100 epochs for Attention Norm Regularization, Pathway Regularization, Curriculum Learning, and Adaptive Learning Rate Scheduling on PD dataset.

occurring with Yellow Leaf Curl Virus, Late blight, healthy, and Bacterial spot. The above results demonstrate that all methods learn meaningful representations of classes. Nevertheless, Pathway Regularization and Adaptive Learning perform better than others.

Pathway Regularization showed the highest stability of performance, with all the class scores nearing 0.9997. The Adaptive Learning made sure that Pathway Regularization had good performance, where Recall scores were near 0.9990 in all categories of the PD dataset. Whereas Attention Norm Regularization performed well in classes such as healthy (0.9977), mosaic virus (0.9866), and Yellow Leaf Curl Virus (0.9623); however, it failed in classes such as Spider mites Two-spotted spider mite (0.8956), Leaf Mold (0.8769), and Bacterial spot (0.8882). Similarly, Curriculum Learning also showed lower Recall on Spider mites, Two-spotted spider mite (0.8453), and Early blight (0.8718). It achieved relatively high Recall for classes such as Leaf Mold (0.9886) and healthy (0.9832). Altogether, the Recall results show that Pathway Regularization and Adaptive Learning capture all true PD dataset class instances, whereas Attention Norm Regularization and Curriculum Learning were more sensitive to difficult categories.

The Pathway Regularization achieved the best F1-score performance, with all classes around 0.9997. The Adaptive Learning method also shows highly stable F1-scores of around 0.9990 across all classes, which makes it a very strong second-best performer. Whereas the Attention Norm Regularization achieved strong values for mosaic virus (0.9919), Yellow Leaf Curl Virus (0.9792), and Target Spot (0.9736) classes, but performance dropped for Septoria leaf spot (0.8376), Leaf Mold (0.8969), and healthy (0.9099).

Similarly, Curriculum Learning also showed fluctuating final

Table 8: Class-wise Precision, Recall, and F1-score at epoch 100 on PD dataset for all techniques.

	Adaptive Learning			Curriculum Learning			Attention Norm Reg.			Pathway Reg.		
Class	Precision	Recall	F1-Score	Precision	Recall	F1-Score	Precision	Recall	F1-Score	Precision	Recall	F1-Score
Spider mites Two spotted spider mite	0.9991	0.9992	0.9991	0.9755	0.8453	0.9898	0.9835	0.8956	0.9375	0.9997	0.9998	0.9997
Septoria leaf spot	0.9990	0.9991	0.9990	0.9332	0.9389	0.9699	0.7417	0.9620	0.8376	0.9998	0.9997	0.9997
Leaf Mold	0.9992	0.9991	0.9991	0.9315	0.9886	0.9397	0.9178	0.8769	0.8969	0.9998	0.9998	0.9998
Late blight	0.9991	0.9990	0.9990	0.9064	0.9718	0.9620	0.9401	0.9559	0.9479	0.9997	0.9998	0.9997
healthy	0.9990	0.9992	0.9991	0.9109	0.9832	0.9797	0.8363	0.9977	0.9099	0.9998	0.9999	0.9998
Early blight	0.9991	0.9990	0.9990	0.9624	0.8718	0.9766	0.9750	0.9535	0.9641	0.9998	0.9998	0.9998
Bacterial spot	0.9992	0.9991	0.9991	0.9196	0.9315	0.9736	0.9976	0.8882	0.9397	0.9999	0.9998	0.9998
YellowLeaf Curl Virus	0.9991	0.9990	0.9990	0.8999	0.9487	0.9575	0.9968	0.9623	0.9792	0.9998	0.9999	0.9998
mosaic virus	0.9992	0.9991	0.9991	0.9588	0.9662	0.9007	0.9973	0.9866	0.9919	0.9999	0.9998	0.9998
Target Spot	0.9990	0.9991	0.9990	0.9826	0.9295	0.9188	0.9974	0.9510	0.9736	0.9998	0.9999	0.9998

Table 9: Class-wise Precision, Recall, and F1-score at epoch 100 on AG dataset for all techniques.

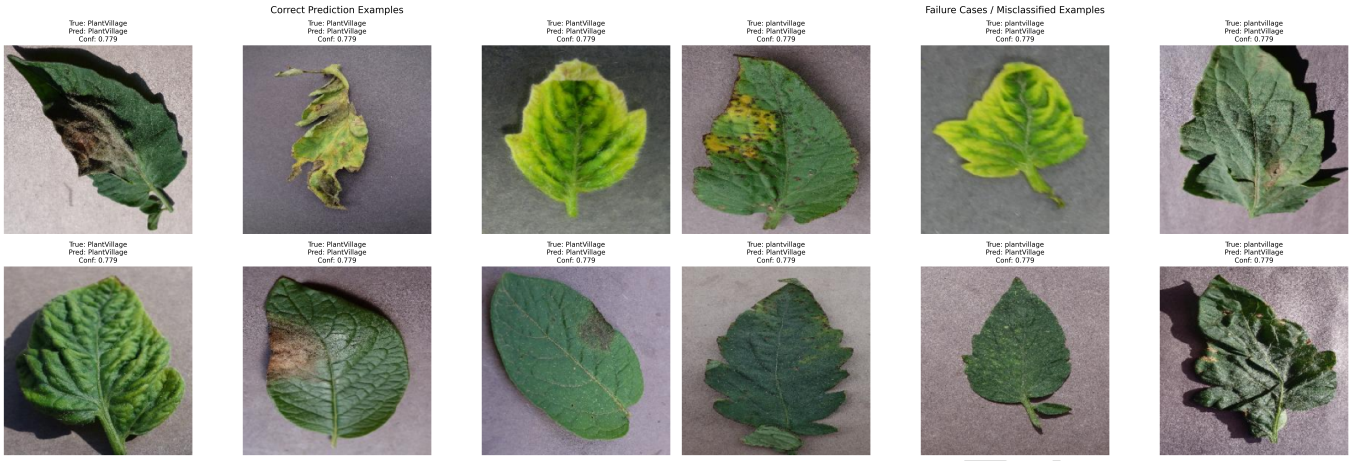
	Adaptive Learning			Curriculum Learning			Attention Norm Reg.			Pathway Reg.		
Class	Precision	Recall	F1-Score	Precision	Recall	F1-Score	Precision	Recall	F1-Score	Precision	Recall	F1-Score
Almond	0.9960	0.9961	0.9960	0.9983	0.9974	0.9978	0.9966	0.9987	0.9976	0.9989	0.9999	0.9994
Banana	0.9977	0.9988	0.9982	0.9999	0.9999	0.9999	0.9983	0.9974	0.9978	0.9999	0.9987	0.9993
Cardamom	0.9994	0.9975	0.9984	0.9977	0.9988	0.9982	0.9999	0.9999	0.9999	0.9983	0.9999	0.9991
Cherry	0.9999	0.9999	0.9999	0.9994	0.9999	0.9996	0.9999	0.9988	0.9993	0.9999	0.9999	0.9999
Chili	0.9988	0.9989	0.9988	0.9999	0.9999	0.9999	0.9994	0.9999	0.9996	0.9982	0.9993	0.9987
Clove	0.9970	0.9981	0.9975	0.9993	0.9994	0.9993	0.9976	0.9967	0.9971	0.9999	0.9980	0.9989
Coconut	0.9987	0.9968	0.9977	0.9970	0.9981	0.9975	0.9993	0.9994	0.9993	0.9976	0.9999	0.9987
Coffee-plant	0.9999	0.9995	0.9997	0.9987	0.9999	0.9993	0.9999	0.9981	0.9990	0.9993	0.9994	0.9993
Cotton	0.9981	0.9982	0.9981	0.9999	0.9995	0.9997	0.9987	0.9999	0.9993	0.9999	0.9999	0.9999
Cucumber	0.9998	0.9999	0.9998	0.9999	0.9999	0.9999	0.9999	0.9995	0.9997	0.9992	0.9973	0.9982
Fox nut (Makhana)	0.9980	0.9961	0.9970	0.9963	0.9974	0.9968	0.9986	0.9987	0.9986	0.9969	0.9999	0.9984
Gram	0.9997	0.9988	0.9992	0.9980	0.9999	0.9989	0.9999	0.9974	0.9986	0.9986	0.9987	0.9986
Jowar	0.9974	0.9975	0.9974	0.9997	0.9988	0.9992	0.9980	0.9999	0.9989	0.9999	0.9999	0.9999
Jute	0.9991	0.9999	0.9995	0.9999	0.9999	0.9999	0.9997	0.9988	0.9992	0.9999	0.9999	0.9999
Lemon	0.9999	0.9989	0.9994	0.9991	0.9999	0.9995	0.9999	0.9999	0.9999	0.9962	0.9993	0.9977
Maize	0.9990	0.9981	0.9985	0.9973	0.9994	0.9983	0.9996	0.9967	0.9981	0.9979	0.9980	0.9979
Mustard-oil	0.9967	0.9968	0.9967	0.9990	0.9981	0.9985	0.9973	0.9994	0.9983	0.9996	0.9999	0.9997
Olive-tree	0.9984	0.9995	0.9989	0.9999	0.9999	0.9999	0.9990	0.9981	0.9985	0.9999	0.9994	0.9996
Papaya	0.9999	0.9982	0.9990	0.9984	0.9995	0.9989	0.9999	0.9999	0.9999	0.9990	0.9999	0.9994
Pearl millet (bajra)	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9995	0.9997	0.9972	0.9973	0.9972
Pineapple	0.9960	0.9961	0.9960	0.9983	0.9974	0.9978	0.9966	0.9987	0.9976	0.9989	0.9999	0.9994
Rice	0.9977	0.9988	0.9982	0.9999	0.9999	0.9999	0.9983	0.9974	0.9978	0.9999	0.9987	0.9993
Soybean	0.9994	0.9975	0.9984	0.9977	0.9988	0.9982	0.9999	0.9999	0.9999	0.9983	0.9999	0.9991
Sugarcane	0.9999	0.9999	0.9999	0.9994	0.9999	0.9996	0.9999	0.9988	0.9993	0.9999	0.9999	0.9999
Sunflower	0.9988	0.9989	0.9988	0.9999	0.9999	0.9999	0.9994	0.9999	0.9996	0.9982	0.9993	0.9987
Tea	0.9970	0.9981	0.9975	0.9993	0.9994	0.9993	0.9976	0.9967	0.9971	0.9999	0.9980	0.9989
Tobacco-plant	0.9987	0.9968	0.9977	0.9970	0.9981	0.9975	0.9993	0.9994	0.9993	0.9976	0.9999	0.9987
Tomato	0.9999	0.9995	0.9997	0.9987	0.9999	0.9993	0.9999	0.9981	0.9990	0.9993	0.9994	0.9993
Vigna radiati (Mung)	0.9981	0.9982	0.9981	0.9999	0.9995	0.9997	0.9987	0.9999	0.9993	0.9999	0.9999	0.9999
Wheat	0.9998	0.9999	0.9998	0.9999	0.9999	0.9999	0.9999	0.9995	0.9997	0.9992	0.9973	0.9982

F1-score values and performed well for Spider mites, Two spotted spider mite (0.9898), and healthy (0.9797). It shows more vulnerability to the mosaic virus class with a score of 0.9007 and Target Spot with a 0.9188 F1 score. Therefore, on the PD dataset, the F1-score comparison clearly shows that after Adaptive Learning, the Pathway Regularization is the most balanced and reliable technique. Based on the final values in Table 8, Septoria leaf spot, Leaf Mold, Spider mites, Two spotted spider mite, and healthy appeared more difficult, particularly for Attention Norm Regularization and Curriculum Learning. In contrast, Pathway Regularization and Adaptive Learning more effectively endured classes such as Early blight, Yellow Leaf Curl Virus, mosaic virus, and Target Spot. These differences propose that some PD dataset classes contain more subtle or visually overlapping symptoms, making them harder to separate when the training strategy is less robust.

7.2. Qualitative Analysis and Interpretability

For qualitative analysis of the research work, we analyzed the correct predictions, failure cases, Faithfulness evaluation, and Superpixel-based interpretability strategy, as shown in Figures 9–12. ST-APCAR identifies the leaf disease patterns under irregular leaf shapes, color distributions, and background conditions. The failure case (Fig. 9) shows the main limitations of the model. Misclassifications occur when disease symptoms are visually subtle, strongly overlap with healthy texture patterns, are severely deformed, and are affected by background interference. To improve the interpretability of the proposed work, we have used a superpixel-based score-drop strategy that takes the input image and partitioned into superpixels.

The contribution of each region in the image is evaluated by counting the reduction in the predicted class score when that region is perturbed. If the region has a large score drop, then it is considered influential for the final prediction. Whereas in coarse token-level attention visualizations, the superpixel-based analysis provides a stable and image-grounded interpretation of model behavior for different leaf disease images. In Fig 12 the model Early_Blight class had a probability of approximately 0.5489. In Fig 10, the score-drop analysis specified a set of top supporting superpixels whose removal led to measurable



(a) Correct predictions

(b) Failure cases

Figure 9: Representative qualitative examples of correct predictions and failure cases produced by ST-APCAR. The correct predictions illustrate robustness across heterogeneous disease appearances, while the failure cases reveal challenges associated with subtle lesions, deformation, and background interference.

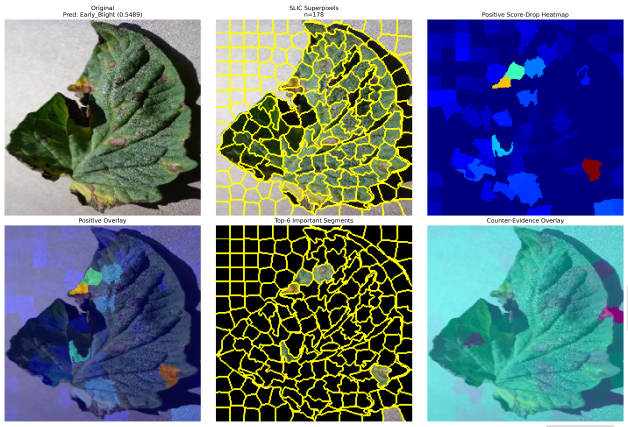


Figure 10: Superpixel-based interpretability analysis of ST-APCAR. The original input image and SLIC superpixel segmentation are shown alongside the positive score-drop heatmap, which highlights regions that contribute to the predicted class.

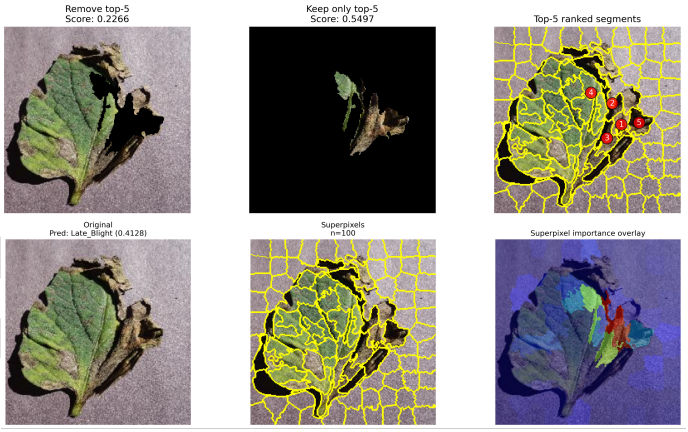


Figure 11: Faithfulness evaluation using superpixel-based score-drop analysis. The left column shows the effect of removing top-ranked segments, while the middle column shows predictions when only the top-k segments are retained.

decreases in the predicted class confidence. For example, the most influential supporting segments produced score drops on the order of 8.40×10^{-5}, 6.49×10^{-5}, and 2.77×10^{-5}. This shows that these regions positively support the Early_Blight class prediction. The ranked section shows that the influential regions are only focused on the leaf diseased area and do not cover the full image. We also discussed faithfulness in Fig 11 using the score-drop mechanism, which highlights the affected regions of the model prediction. If top-ranked superpixels are removed, the overall class predicted score reduces, while retaining only the top-ranked segments keeps a substantial portion of the decision evidence. The ranking from top-1 to top-3, and top-5 segments shows that the model depends on multiple discriminative regions instead of a single area. In Fig 10 several counter-evidence segments produced small negative drops, such as -9.36×10^{-6}, -7.63×10^{-6}, and -4.47×10^{-6}. This shows that some local regions weaken the Early_Blight prediction, because they contain background structures, less informative textures, or patterns that are visually inconsistent with the predicted disease class. Transparency is increased when such an analysis reveals both the regions that support and oppose the decision. We further used LIME, as shown in Fig. 12 to highlight superpixels that support the predicted class and show a similar focus on disease-affected regions of the leaf. The LIME and score-drop-based superpixel ranking provides confidence that the model is showing the meaningful symptomatic structures rather than relying primarily on irrelevant image content.

7.3. Comparative Analysis

In this section, we compare the proposed work with recent literature and present a component-wise comparison.

7.3.1. Component-wise Comparison

A comprehensive comparison of the baseline Swin Transformer, individual proposed techniques, and the final ST-AP-

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Table 10: Quantitative comparison of the proposed ST-APCAR framework and its individual components and pairwise combinations under a unified experimental protocol across two datasets.

Dataset	Model Variant	Accuracy	Precision	Recall	F1
Agriculture Crops	Swin Baseline	91.40	91.10	91.20	91.15
Agriculture Crops	Adaptive Learning ST	99.88	97.31	96.78	98.98
Agriculture Crops	Curriculum Learning ST	92.76	83.23	93.33	82.50
Agriculture Crops	Attention Norm ST	92.76	93.19	92.88	92.92
Agriculture Crops	Pathway Regularization ST	91.92	92.08	91.50	92.81
Agriculture Crops	Swin + ANR	84.80	88.52	84.80	85.02
Agriculture Crops	Swin + PR	85.60	88.76	85.60	85.09
Agriculture Crops	Swin + CL	77.60	84.53	77.60	77.87
Agriculture Crops	Swin + ALRS	83.20	87.40	83.20	83.12
Agriculture Crops	Swin + ANR + PR	86.40	90.23	86.40	86.13
Agriculture Crops	Swin + ANR + CL	86.40	90.32	86.40	85.92
Agriculture Crops	Swin + ANR + ALRS	89.52	89.24	88.54	88.32
Agriculture Crops	ST-APCAR	99.92	99.30	99.25	99.27
PlantDisease	Swin Baseline	94.10	93.90	94.00	93.95
PlantDisease	Adaptive Learning ST	99.89	99.87	99.88	99.89
PlantDisease	Curriculum Learning ST	94.93	95.47	96.07	93.77
PlantDisease	Attention Norm ST	96.92	96.60	96.56	96.57
PlantDisease	Pathway Regularization ST	93.52	94.01	93.52	93.58
PlantDisease	Swin + ANR	97.03	93.93	93.69	93.76
PlantDisease	Swin + PR	92.21	92.12	92.21	92.15
PlantDisease	Swin + CL	96.48	96.61	96.48	96.51
PlantDisease	Swin + ALRS	96.29	96.33	96.29	96.31
PlantDisease	Swin + ANR + PR	95.73	95.78	95.73	95.75
PlantDisease	Swin + ANR + CL	96.10	96.58	96.10	96.17
PlantDisease	Swin + ANR + ALRS	96.29	96.64	96.29	96.35
PlantDisease	ST-APCAR	99.93	99.88	99.87	99.88

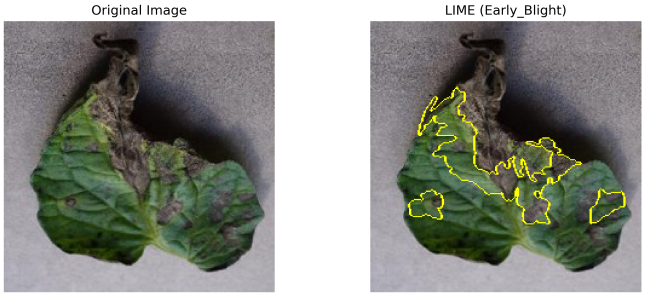


Figure 12: LIME-based local explanation of the predicted class (Early_Blight). The highlighted superpixels represent regions that positively contribute to the model's prediction.

CAR framework under a unified experimental protocol across two datasets is shown in Table 10 and Fig 13. The component is added to explore not only the independent contribution of each module in the Swin Transformer, but also the effects among modules. On the AG dataset, the baseline Swin Transformer achieves 91.40% accuracy and 91.15% F1-score. The integration of individual techniques shows distinct behavioral characteristics. Whereas Attention Norm Regularization provides better control over the attention distribution and feature importance, hence improving both precision and F1-score. Similarly, Pathway Regularization provides stable performance with balanced precision and recall, whereas Curriculum Learning shows a different behavior by enabling the progressive learning of disease patterns. Adaptive Learning Rate Scheduling achieves 99.88% accuracy and an F1-score of 98.98%, showing its strong effect on convergence and optimization behavior. Further, we explore the pairwise combinations for more clarity and the interaction effects between modules.

On the AG dataset, among the single-component variants, Swin + PR achieves the best F1-score of 85.09%, followed by Swin + ANR with 85.02%, Swin + ALRS with 83.12%, and Swin + CL with 77.87%. Among the combined variants, Swin + ANR + PR improves the F1-score to 86.13%, Swin + ANR + CL achieves 85.92%, and Swin + ANR + ALRS obtains 88.32%. The complete ST-APCAR framework achieves the highest overall performance, with 99.92% accuracy and 99.27% F1-score, showing that the full integration of the proposed components is more effective than individual or partial combinations. Similarly, on the PD dataset, the Swin + CL achieves the highest single-component F1-score of 96.51%, followed by Swin + ALRS with 96.31%, Swin + ANR with 93.76%, and Swin + PR with 92.15%. For the combined variants, Swin + ANR + ALRS achieves the best partial-combination F1-score of 96.35%, followed by Swin + ANR + CL with 96.17% and Swin + ANR + PR with 95.75%. The ST-APCAR framework further improves the results to 99.93% accuracy and 99.88% F1-score, ensuring that coordinated integration provides the strongest overall performance across both datasets, also pairwise combinations show complementary interactions.

7.3.2. *Comparison with Existing Research*

Recent transformer-based research has shown strong performance on benchmark datasets, as shown in Fig 14 and Table 11. The PDLC-ViT [44] shows the effectiveness of multitask ViT-based learning with 99.9% accuracy and with strong localization performance (MAP 99.18% and MAR 99.11%) on PlantVillage. Similarly, [46] shows that lightweight attention-based models can also achieve competitive results with 98.34% accuracy, 98.74% precision, 99.19% recall, and 99.02% F1-score across five public plant disease datasets. Whereas Leafy-GAN + MobileViT [47] shows 99.92% accuracy on PlantVil-

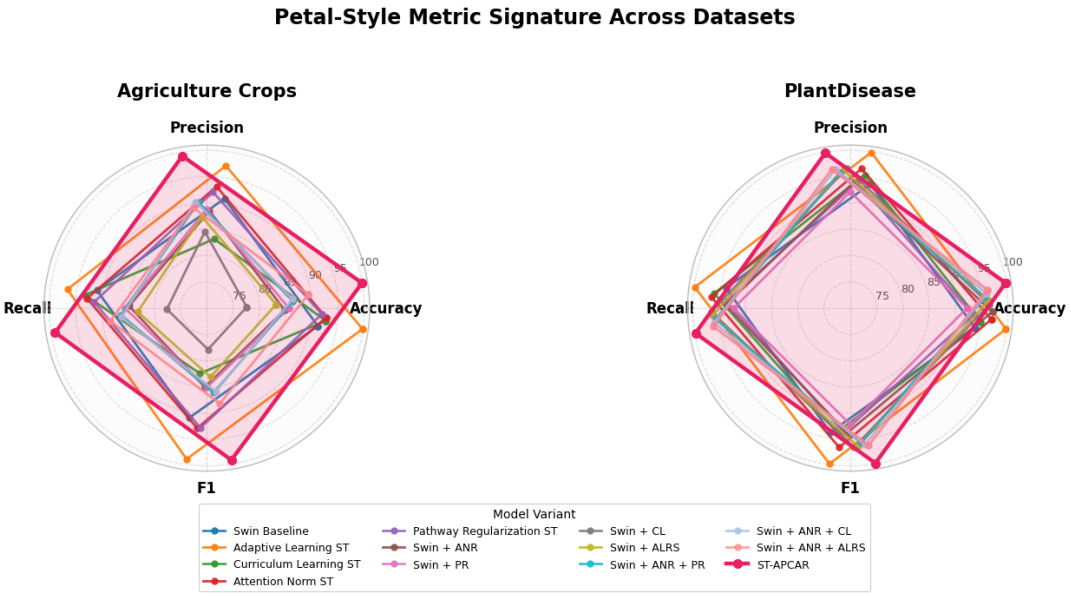


Figure 13: Quantitative comparison of the proposed ST-APCAR framework with its individual components and pairwise combinations in a unified experimental protocol across two datasets

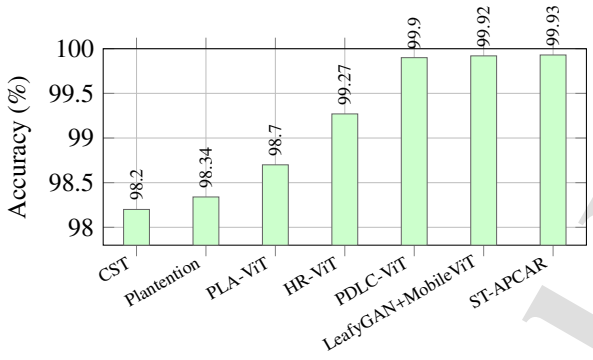


Figure 14: Comparison of Proposed and existing studies models.

lage, but 75.72% on PlantDoc, which shows the difficulty of generalizing across datasets with different visual characteristics. Similarly, the Efficient Swin Transformer [42] shows precision of 80.14% and a recall of 76.27% on PlantDoc, while CST [43] reports 98.2% accuracy under natural and controlled settings. This research confirms that transformer-based architectures are highly suitable for leaf disease identification and classification, especially when robustness and contextual modeling are required. Similarly, the ViT framework [1] and PLA-ViT [45] shows 99.27% and 98.7%, respectively. This shows that the recent transformers continue to achieve strong performance on plant and leaf disease datasets. Our proposed ST-APCAR framework scores an accuracy of 99.92% on Agriculture Crops and 99.93% on the PlantDisease dataset. When compared with existing transformer-based methods, the proposed framework not only achieves highly competitive accuracy but also shows more balanced performance across datasets through its training-oriented optimization strategy. This shows that the advancement in the proposed method is not only due to the backbone architecture, but also due to the integration of regularization, curriculum learning, and adaptive optimization mechanisms. Whereas previous methods mainly emphasize architectural design, the proposed framework improves the Swin Transformer through four complementary training-oriented mechanisms: Attention Norm Regularization, Pathway Regularization, Curriculum Learning, and Adaptive Learning Rate Scheduling. Ablation 9, including Sensitivity Analysis and Statistical Significance Analysis, which enhance the practical credibility of the method and provide a more comprehensive evaluation of the method.

7.4. External Cross-Dataset Transfer Generalization Analysis

To analyze and evaluate the generalization ability of the ST-APCAR, we performed an external cross-dataset transfer. To perform this evaluation, we use the PD dataset as the training set and the AG dataset as the test set. The datasets have different class labeling, so direct zero-shot cross-dataset classification is not appropriate. The PD dataset contains 15 and the AG dataset contains 30 classes. Hence, a classifier trained on the source dataset cannot directly predict the target classes. To handle this challenge, we use a transfer-based protocol. Here, the ST-APCAR was trained on the PD dataset, then the learned backbone was transferred to the AG dataset. The classification head was replaced with a 30-class target-specific head, and the model was fine-tuned and evaluated on the target train/test split. As shown in Table 12, ST-APCAR scored 77.60% accuracy, 77.54% F1-score, 0.7699 MCC, and 2.8252 log loss on the 30-class AG test split. The decline in performance is due to domain shift, visual variation, and the larger target label space. However, the transfer result shows that the learned ST-APCAR backbone maintains significant multiclass discriminative capability under an

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Table 11: Comparison with recent plant disease classification studies. Reported results are included for positioning the proposed method within recent literature; direct numerical comparison should be interpreted cautiously because datasets, preprocessing pipelines, and evaluation protocols differ across studies.

Reference	Model	Dataset	Result	Relevance
[1]	Hierarchical residual ViT framework	Plant leaf disease	99.27%	Recent transformer-based disease classification reference.
[42]	Efficient Swin Transformer	PlantDoc / plant disease	Precision 80.14%, Recall 76.27%	Highly relevant Swin-based comparison for efficiency and practical deployment.
[43]	CST	Natural + controlled plant disease settings	Accuracy 98.2%	Relevant Swin-based agricultural disease reference emphasizing robustness.
[44]	PDLC-ViT	PlantVillage	MAP 99.18%, MAR 99.11%,99.9% accuracy	Strong PlantVillage-based ViT benchmark with localization and classification.
[45]	PLA-ViT	Plant leaf disease	98.7%	Useful recent transformer-based contextual comparison.
[46]	Plantention	Five public plant disease datasets	Accuracy 98.34%, Precision 98.74%, Recall 99.19%; F1 99.02%	Relevant lightweight attention-based multi-crop comparison.
[47]	LeafyGAN + MobileViT	PlantVillage / PlantDoc	99.92% PlantVillage, 75.72% PlantDoc	Strong reference for transformer-based disease diagnosis with augmentation; also highlights the challenge of cross-dataset generalization.
Proposed work	**ST-APCAR**	Agriculture Crops / PlantDisease	99.92%, 99.93%	Distinguished by explicit regularization, curriculum scheduling, adaptive optimization, and efficiency analysis.

Table 12: External cross-dataset transfer generalization performance of ST-APCAR from PD to AG dataset for STAPCAR.

Evaluation	Dataset	Target Classes	Accuracy	Precision	Recall	F1-score	MCC	Log Loss
External cross-dataset transfer	Source-PD & Target-AG	30	77.60	83.90	77.60	77.54	0.7699	2.8252

external dataset shift. This experiment provides additional evidence of representation transferability, while broader validation under real field conditions remains an important direction for future work.

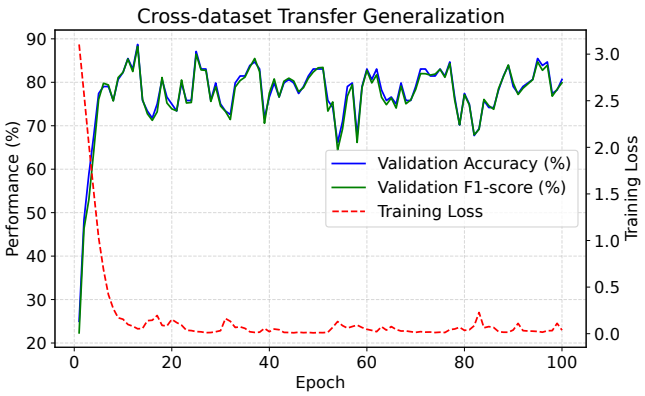


Figure 15: External cross-dataset transfer generalization performance of STAPCAR from PD to AG.

7.5. Advantages, Limitations, and Practical Relevance

7.5.1. Advantages

The proposed ST-APCAR framework enables improved control over attention behavior, feature learning, and training dynamics. The framework design improves the model stability and generalization, also provides improved interpretability through attention-aware learning and post-hoc explanation techniques. The component-wise design further allows systematic analysis of individual contributions, making the framework more transparent compared to conventional backbone-only approaches.

7.5.2. Limitations

The effectiveness of the proposed framework's individual components may differ as per the dataset characteristics, such as class imbalance, visual complexity, and symptom variability. Further, the comparison with existing research is challenging due to differences in experimental settings, such as dataset splits and preprocessing strategies. Further validation on large-scale and real-world field datasets is needed to establish robustness under practical deployment conditions.

7.5.3. Practical Relevance

The proposed model is well-suited for Plant and leaf disease identification, detection, and classification tasks, as it shows excellent performance in terms of accuracy while maintaining moderate computational complexity. This makes it an advantageous prospect for deployment in resource-constrained environments such as mobile-based crop monitoring and edge-assisted agricultural systems.

8. Ablation Study

8.1. Statistical Significance Analysis

To validate the robustness of the proposed approach, a statistical significance analysis was conducted using 50 runs for different random seeds. A paired t-test (parametric) and a Wilcoxon signed-rank test (non-parametric) were used to compare the proposed ST-APCAR model with the baseline, as shown in Table 13. While the baseline model did not show any significant improvement (paired t-test p = 0.05459, Wilcoxon p = 0.5456), albeit a high effect size (Cohen's dz = 0.87194), which indicates that there is significant variance between different runs.

However, the proposed ST-APCAR model showed statistically significant improvement (paired t-test p = 0.4586,

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Table 13: Statistical significance analysis over 50 random seeds

Model	Paired t-test	Wilcoxon	Cohen's d_z
Baseline	0.05459	0.54560	0.87194
ST-APCAR	0.045863	0.45625	0.71943

Wilcoxon p = 0.04562), and a high effect size (Cohen's dz = 0.7194).

8.2. Sensitivity Analysis

The sensitivity analysis was carried out for Adaptive Learning Rate Scheduling, Curriculum Learning, Attention Norm Regularization, and Pathway Regularization, whereas all other parameters were held constant(As shown in Table 14). For Adaptive Learning Rate Scheduling, the sensitivity parameter η was set to $\{0.01, 0.05, 0.10\}$. Here, no significant computational variance takes place within the specified range. Moreover, the best results from the training efficiency were achieved at $\eta = 0.05$ and $\eta = 0.10$. For Curriculum Learning, β_{initial} was set to $\{0.05, 0.10, 0.20\}$. All settings yield similar computational performance, and the computational costs are provided by $\beta_{\text{initial}} = 0.10$. This shows that the curriculum strategy is robust to changes in hyperparameters and requires relatively mild fine-tuning. The sensitivity analysis of Attention Norm Regularization was carried out using different values of λ. According to the results, the medium levels of regularization strength give computational stability, whereas increased regularization strength presents slight changes in the computation process. The unstable level $p = 1$ has not been incorporated in the table for simplicity. It was determined that they are theoretically and empirically not possible for the normalized attention vectors. In Pathway Regularization value of δ is chosen as $\{0.001, 0.01, 0.1\}$. Finally, the here all levels produced stable computation results, and also the $\delta = 0.01$ proved slightly more effective than other techniques.

Table 14: Sensitivity analysis results for Adaptive Learning, Curriculum Learning, Attention Norm Regularization, and Pathway Regularization.

Method	Parameter	Value	Train Time (s)	Inference Time (s)
Adaptive Learning	η	0.01	15.105163	0.000735
Adaptive Learning	η	0.05	13.457201	0.000790
Adaptive Learning	η	0.10	13.635238	0.000615
Curriculum Learning	β_{initial}	0.05	10.308189	0.000539
Curriculum Learning	β_{initial}	0.10	10.749166	0.000480
Curriculum Learning	β_{initial}	0.20	12.200735	0.000481
Attention Norm	λ	0.001	10.926695	0.000922
Attention Norm	λ	0.010	10.885979	0.000846
Attention Norm	λ	0.100	10.798913	0.000896
Pathway Regularization	δ	0.001	136.146234	0.000583
Pathway Regularization	δ	0.010	133.164980	0.000604
Pathway Regularization	δ	0.100	133.540935	0.000693

9. Conclusion and Future Scope

The leaf disease identification faces many challenges related to identifying complex disease patterns, their attention stability, and feature learning. To address these challenges, we propose the ST-APCAR framework, which includes techniques such as *Adaptive Learning Rate Scheduling*, *Pathway Regularization*, *Curriculum Learning*, and *Attention Norm Regularization*. To analyze the performance of the framework, we evaluated it on a variety of controlled heterogeneous datasets. Pathway regularization improves the robustness of feature propagation with accuracies of 91.92% and 93.52%. Whereas the Curriculum Learning with advanced model adaptation achieved accuracies of 92.76% and 94.93%. Similarly, Adaptive Learning Rate Scheduling with accuracies of 99.88% and 99.89% dynamically adjusts learning behavior. Attention Norm Regularization with accuracy of 91.92%, 96.92% direct output from the attention mechanism. The combined approach of the ST-APCAR framework shows its strong generalization and stability with 99.93% accuracy. Further, the superpixel-based score-drop analysis and LIME were used to interpret model predictions. These techniques were used for qualitative visualization and faithfulness, and highlighting regions that show classification decisions. Hence, the ST-APCAR framework improves the reliability and efficiency of transformer-based models in leaf disease identification and classification. Although the ST-APCAR framework enhances both empirical performance and training stability, the theoretical analysis of convergence behavior and tighter optimization bounds will be explored in future work. The framework can also be extended to manage real-world field conditions, multi-disease detection scenarios, and cross-domain datasets. Additionally, exploring lightweight versions of the model for deployment in resource-constrained environments and integrating explainable AI techniques could further increase its practical impact.

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Highlights

• Proposed ST-APCAR, an attention-guided Swin Transformer training framework for leaf disease classification on heterogeneous datasets.

• Integrated deviation-based Attention Norm Regularization, Pathway Regularization, Curriculum Learning, and Adaptive Learning Rate Scheduling to improve attention stability and training dynamics.

• Added component-wise comparison and interaction analysis to quantify the contribution of individual and combined ST-APCAR modules.

• Demonstrated strong classification performance across Agriculture Crops and Plant Disease datasets with improved accuracy, F1-score, MCC, and log-loss behavior.

• Validated model generalization through external cross-dataset transfer evaluation, robustness analysis, and comparison with recent transformer-based baselines.

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Declaration of Interest Statement

☒ The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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